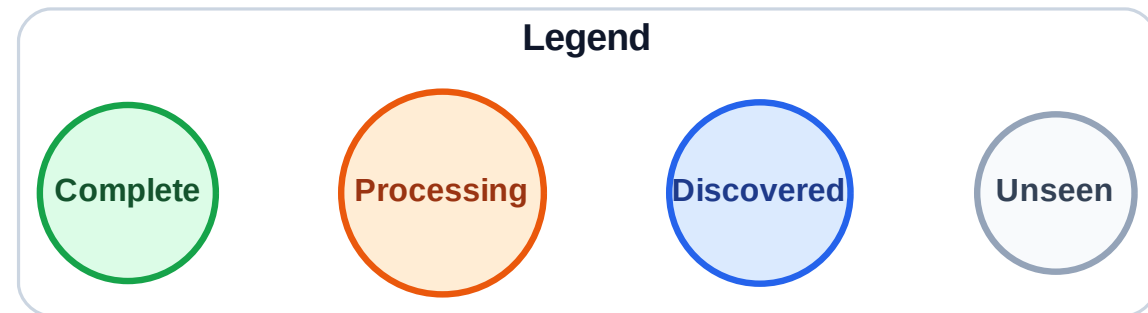
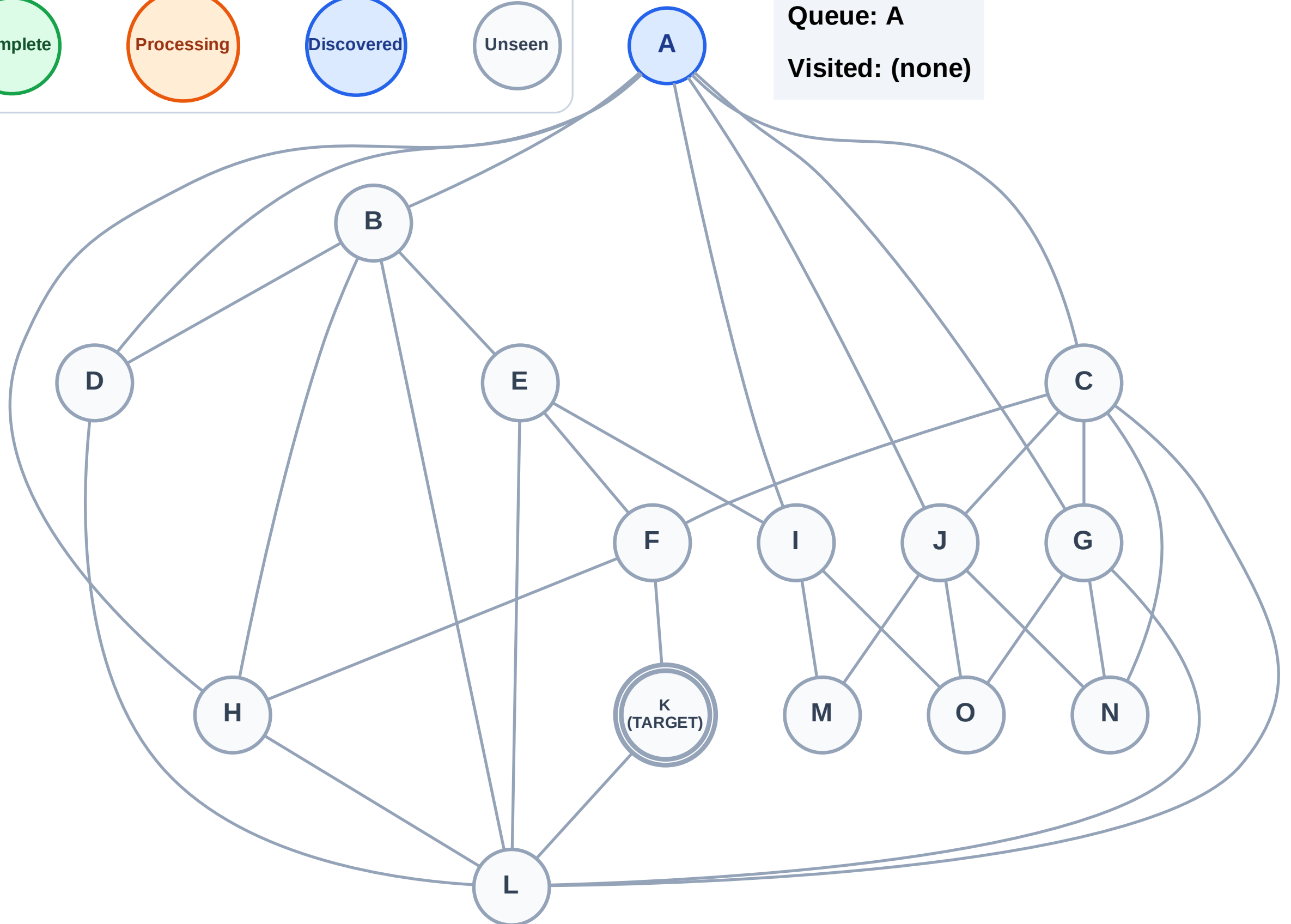


BFS Traversal

Step 1: Start at A (Seeking target node K)



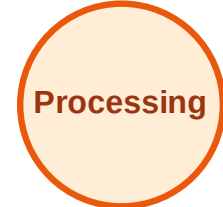
Queue: A
Visited: (none)



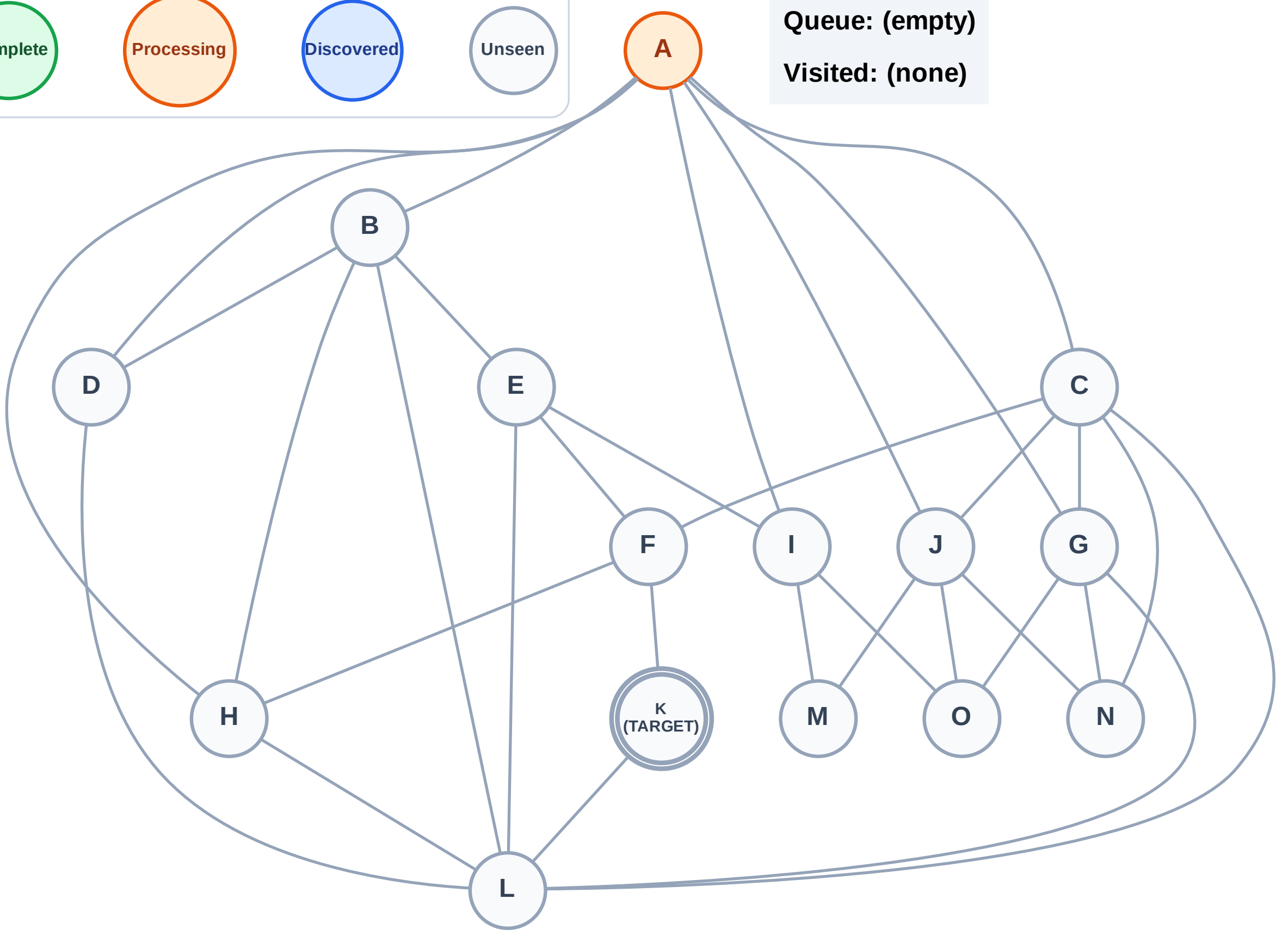
BFS Traversal

Step 2: Process node A

Legend



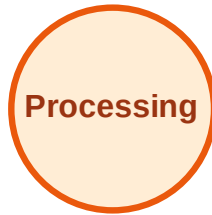
Queue: (empty)
Visited: (none)



BFS Traversal

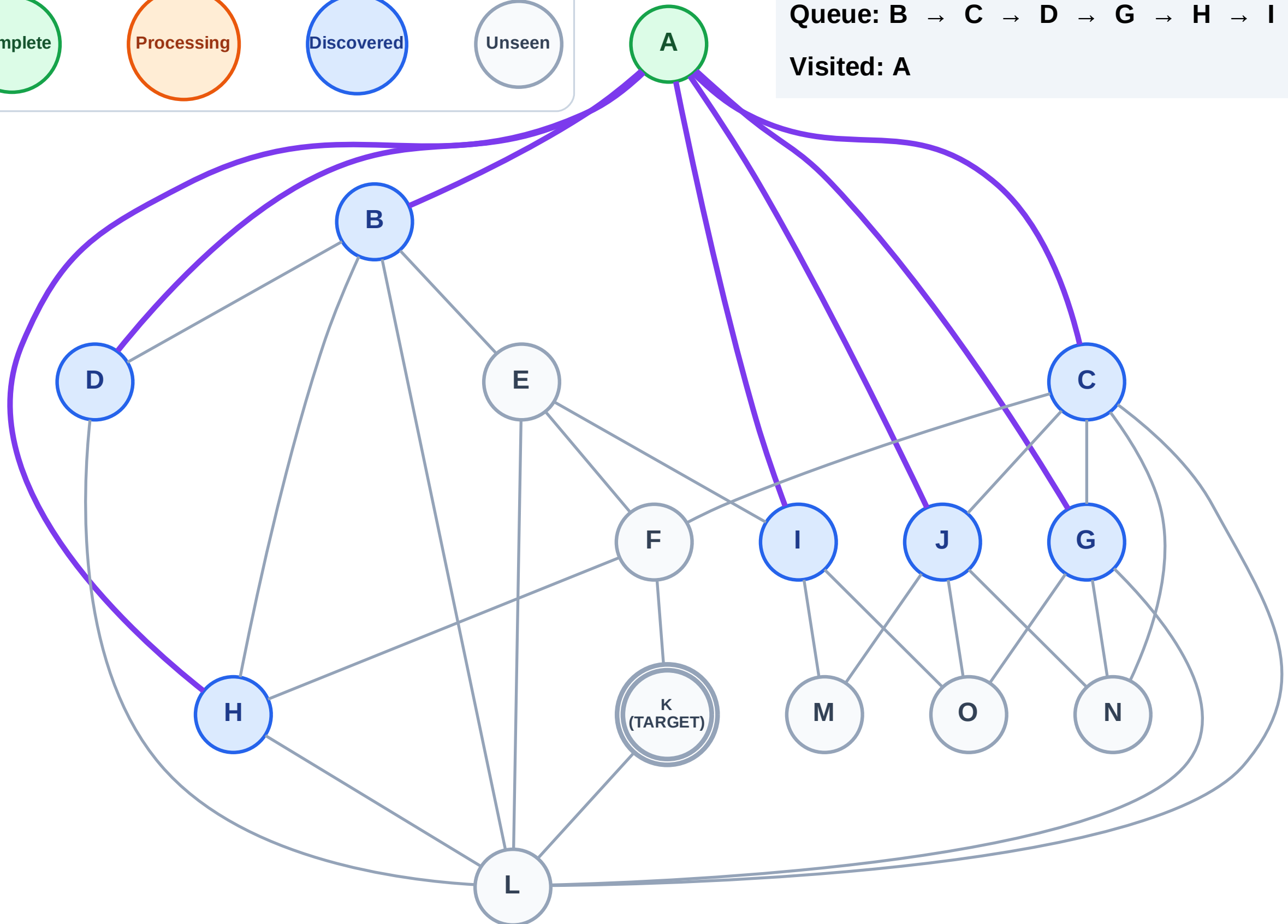
Step 3: Discover B, C, D, G, H, I, J from A

Legend



Queue: B → C → D → G → H → I → J

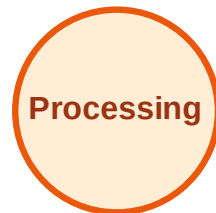
Visited: A



BFS Traversal

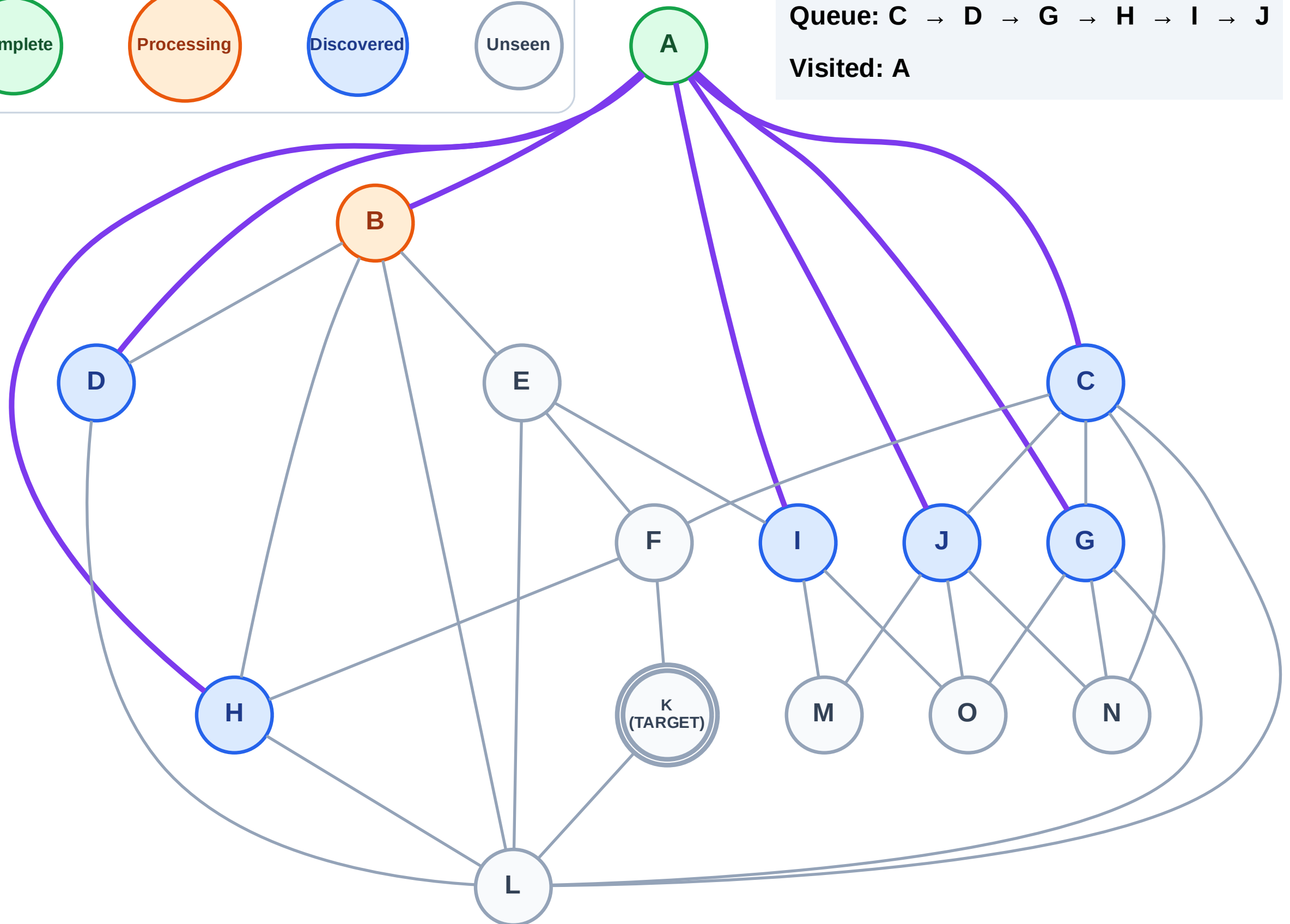
Step 4: Process node B

Legend



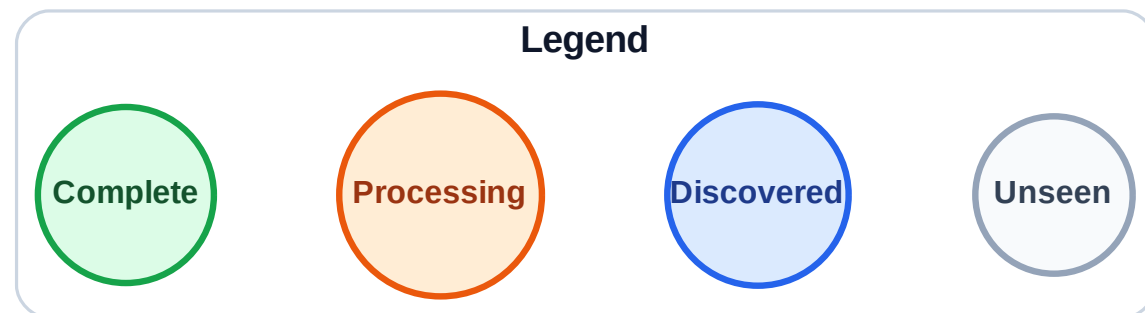
Queue: C → D → G → H → I → J

Visited: A



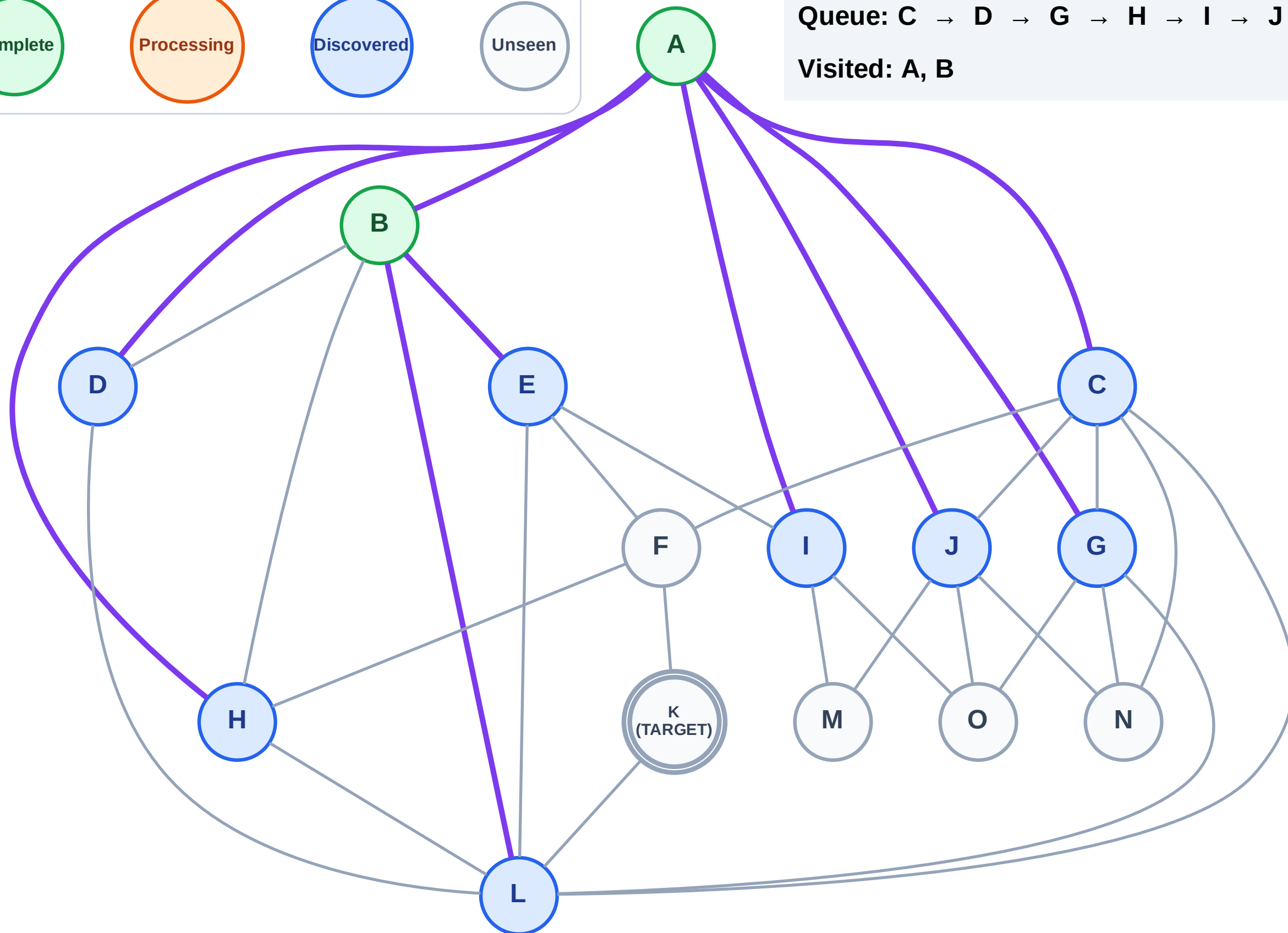
BFS Traversal

Step 5: Discover E, L from B



Queue: C → D → G → H → I → J → E → L

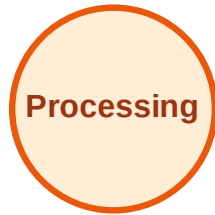
Visited: A, B



BFS Traversal

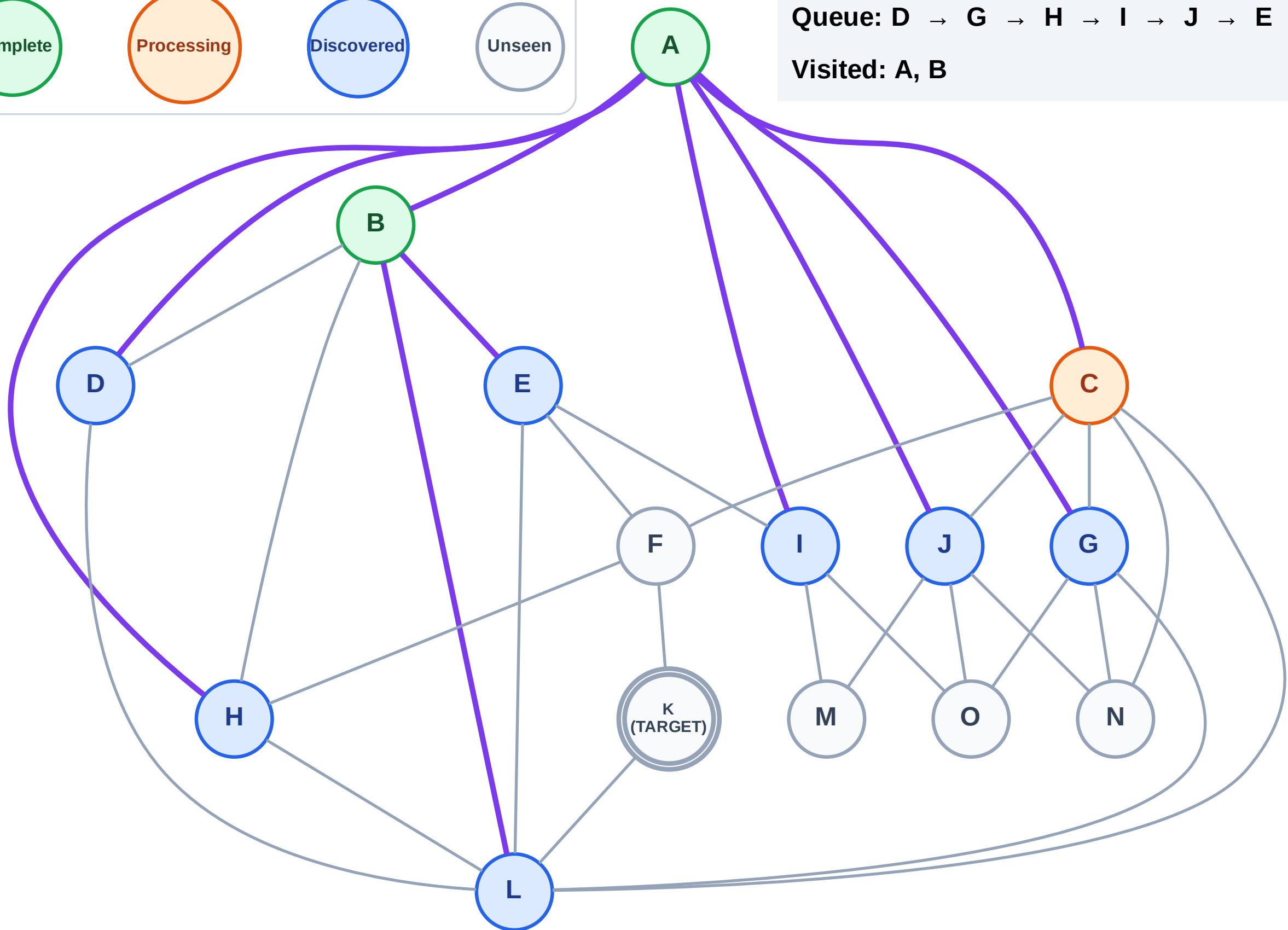
Step 6: Process node C

Legend



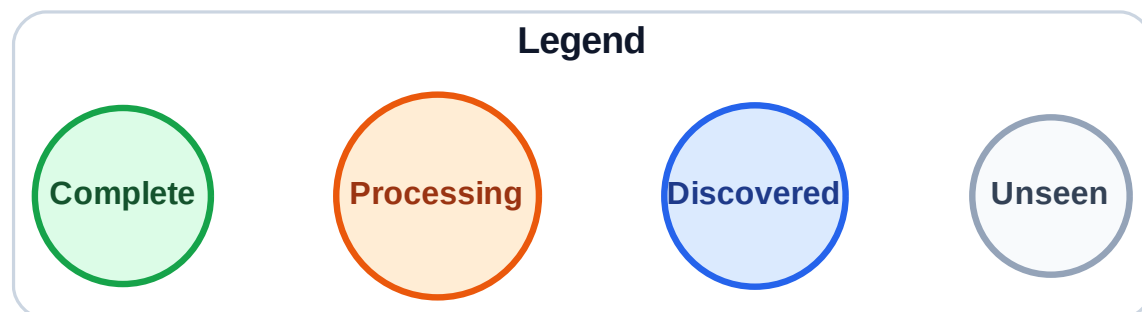
Queue: D → G → H → I → J → E → L

Visited: A, B



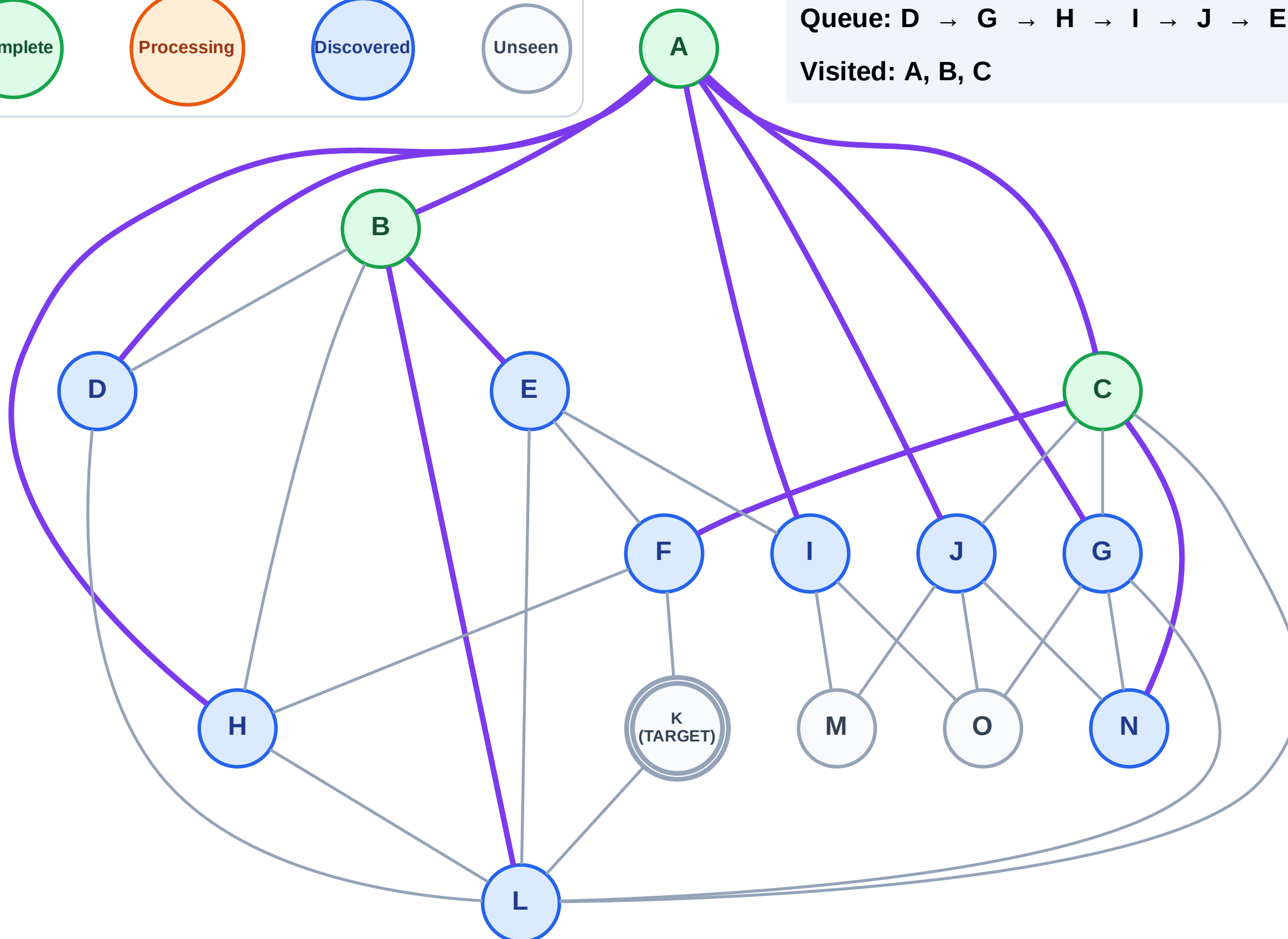
BFS Traversal

Step 7: Discover F, N from C



Queue: D → G → H → I → J → E → L → F → N

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

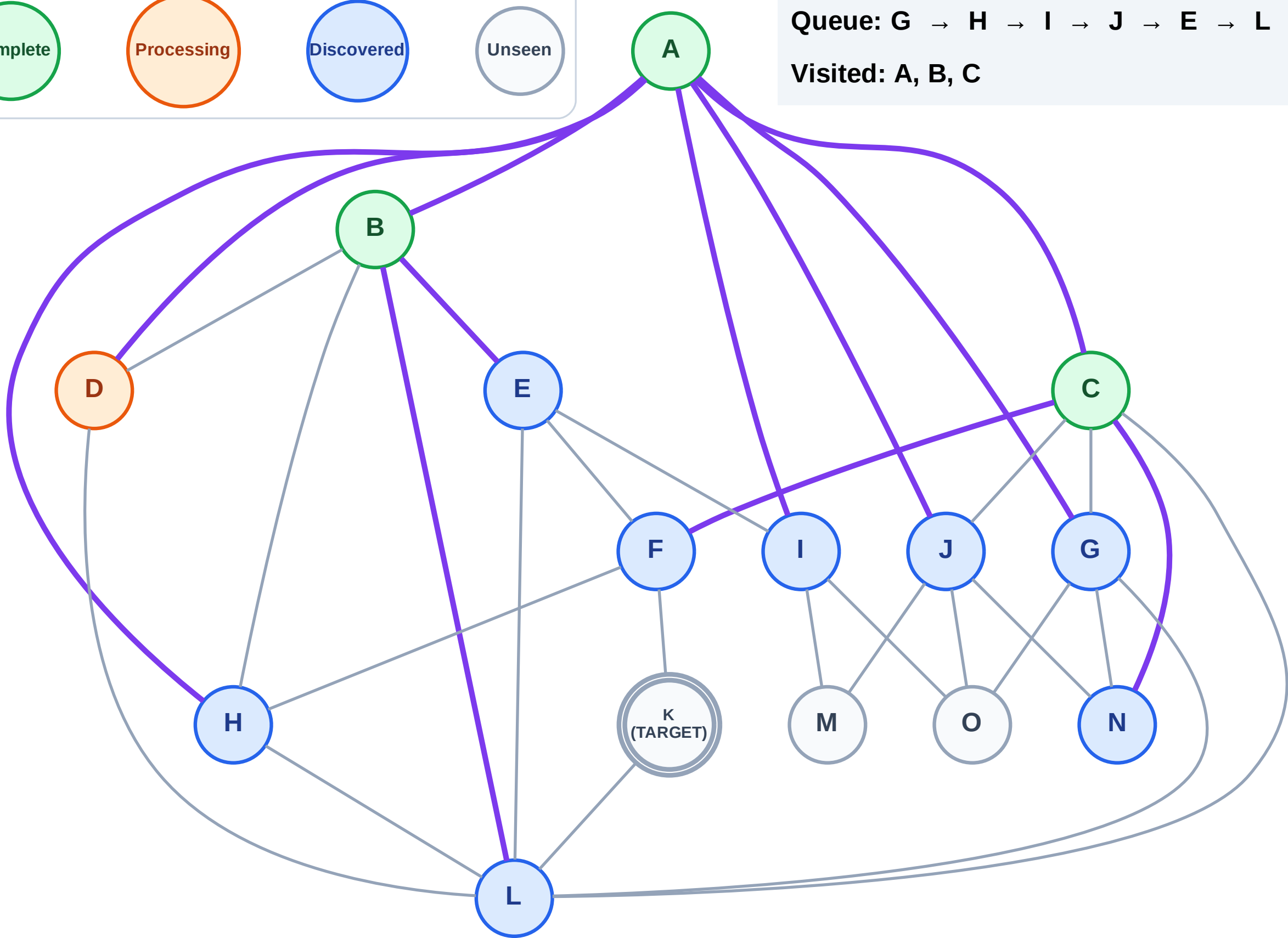
Processing

Discovered

Unseen

Queue: G → H → I → J → E → L → F → N

Visited: A, B, C



BFS Traversal

Step 9: No new neighbors from D

Legend

Complete

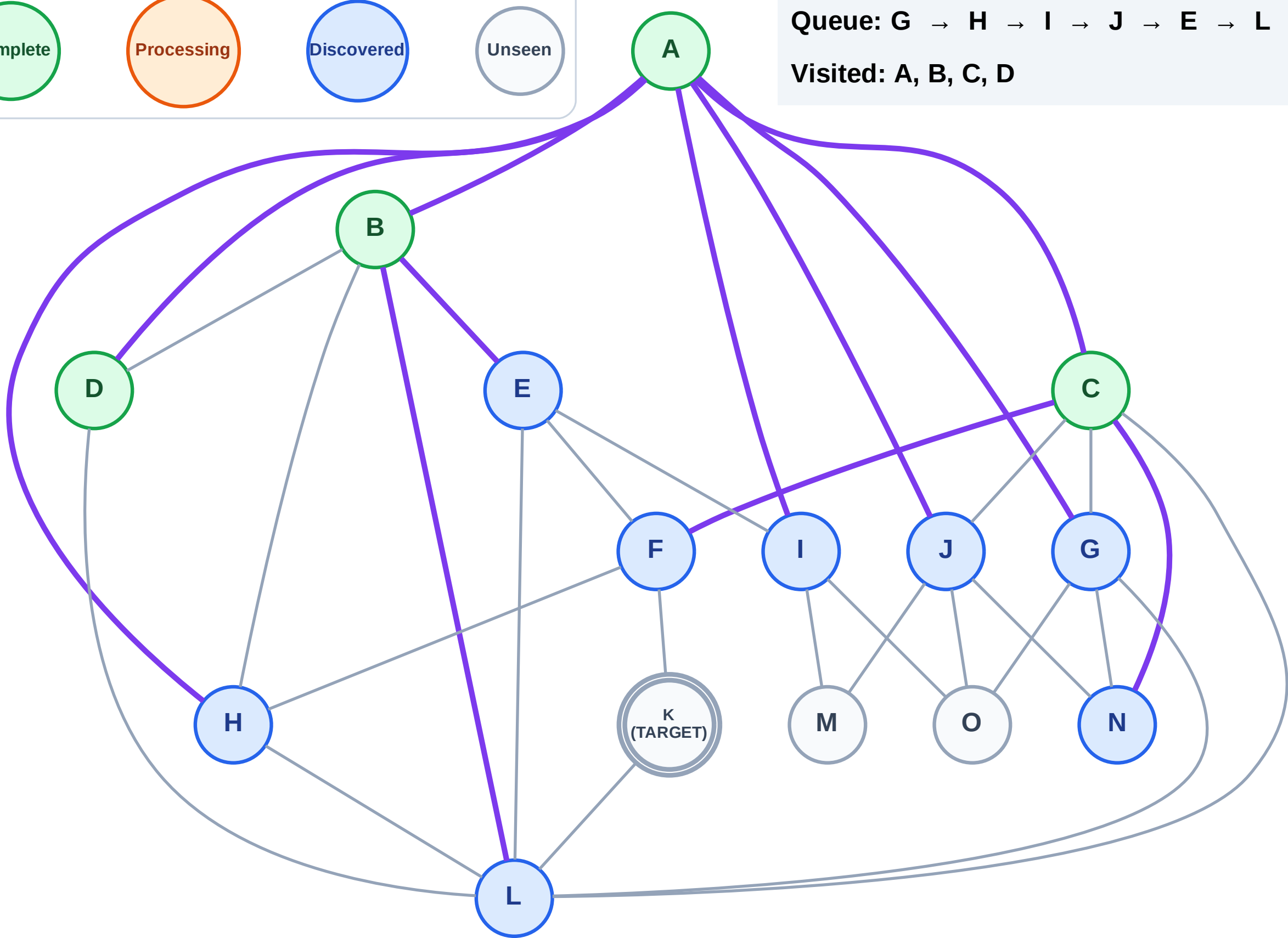
Processing

Discovered

Unseen

Queue: G → H → I → J → E → L → F → N

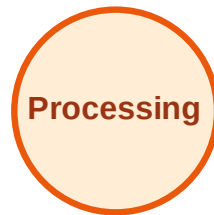
Visited: A, B, C, D



BFS Traversal

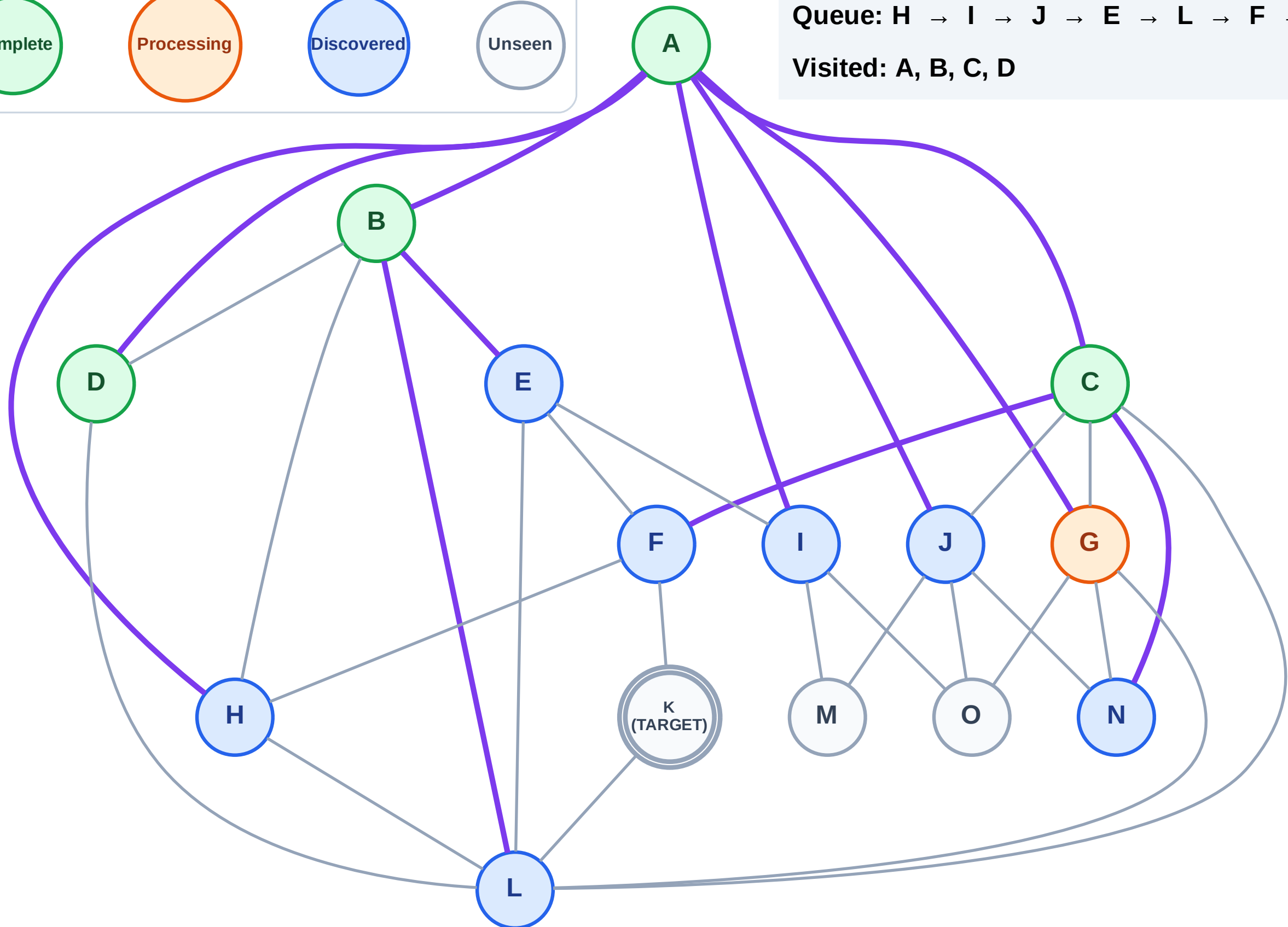
Step 10: Process node G

Legend



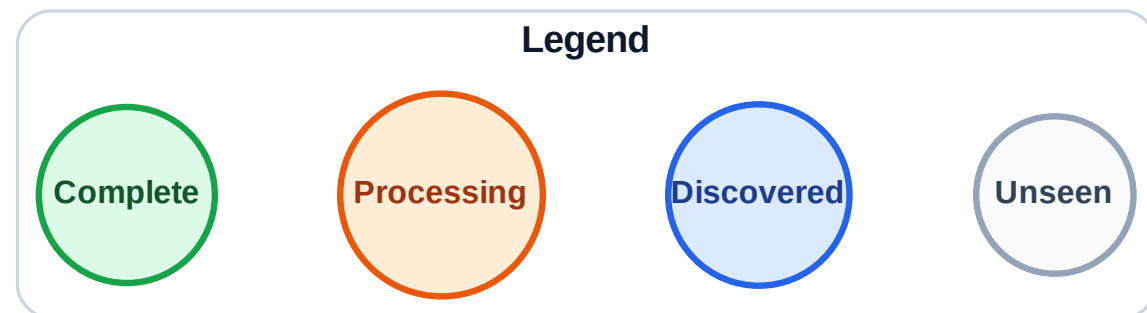
Queue: H → I → J → E → L → F → N

Visited: A, B, C, D



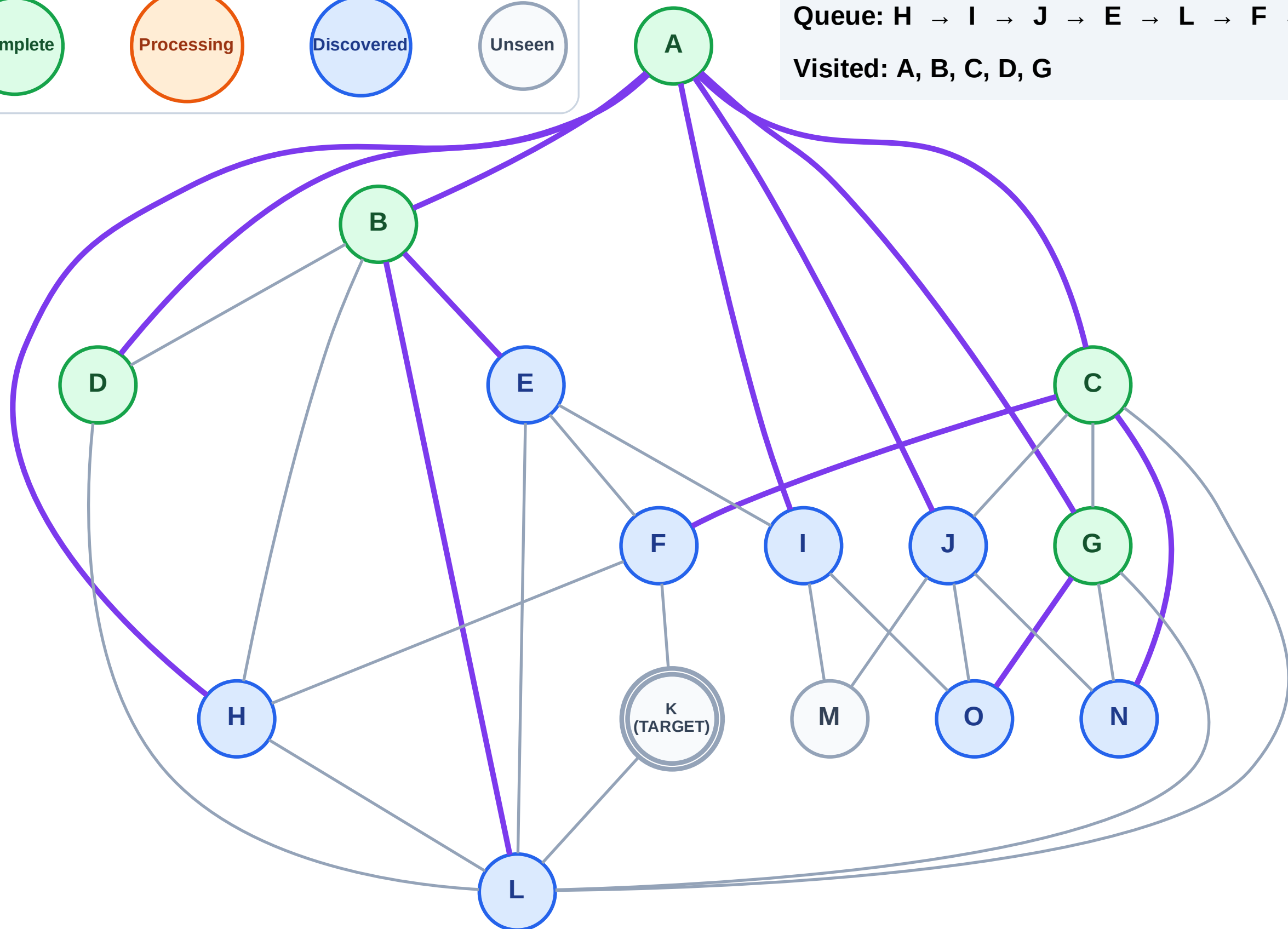
BFS Traversal

Step 11: Discover O from G



Queue: H → I → J → E → L → F → N → O

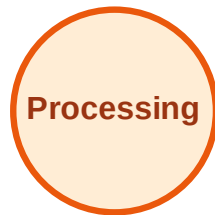
Visited: A, B, C, D, G



BFS Traversal

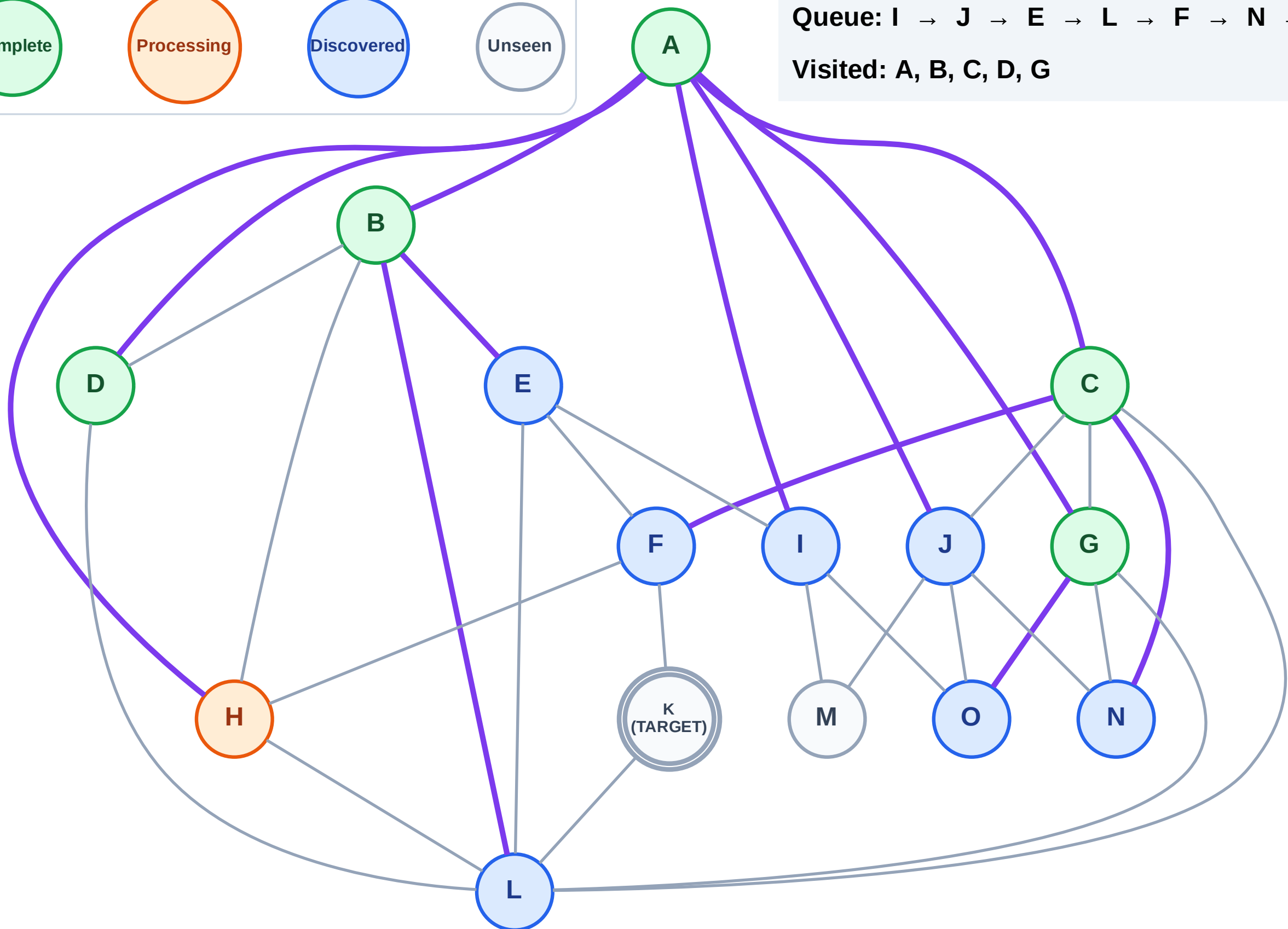
Step 12: Process node H

Legend



Queue: I → J → E → L → F → N → O

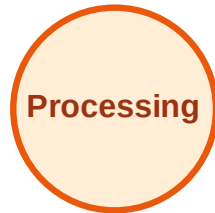
Visited: A, B, C, D, G



BFS Traversal

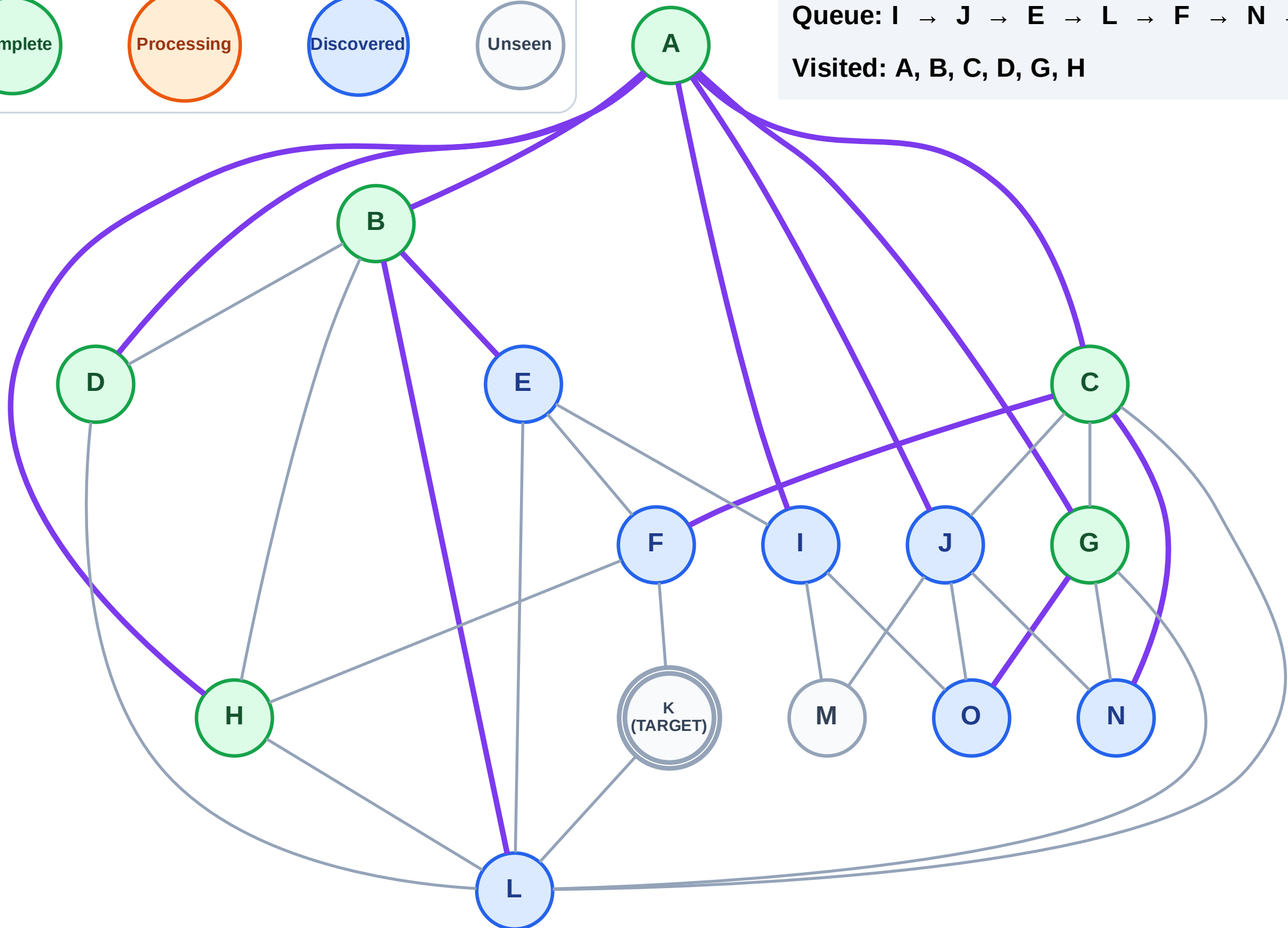
Step 13: No new neighbors from H

Legend



Queue: I → J → E → L → F → N → O

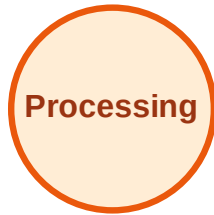
Visited: A, B, C, D, G, H



BFS Traversal

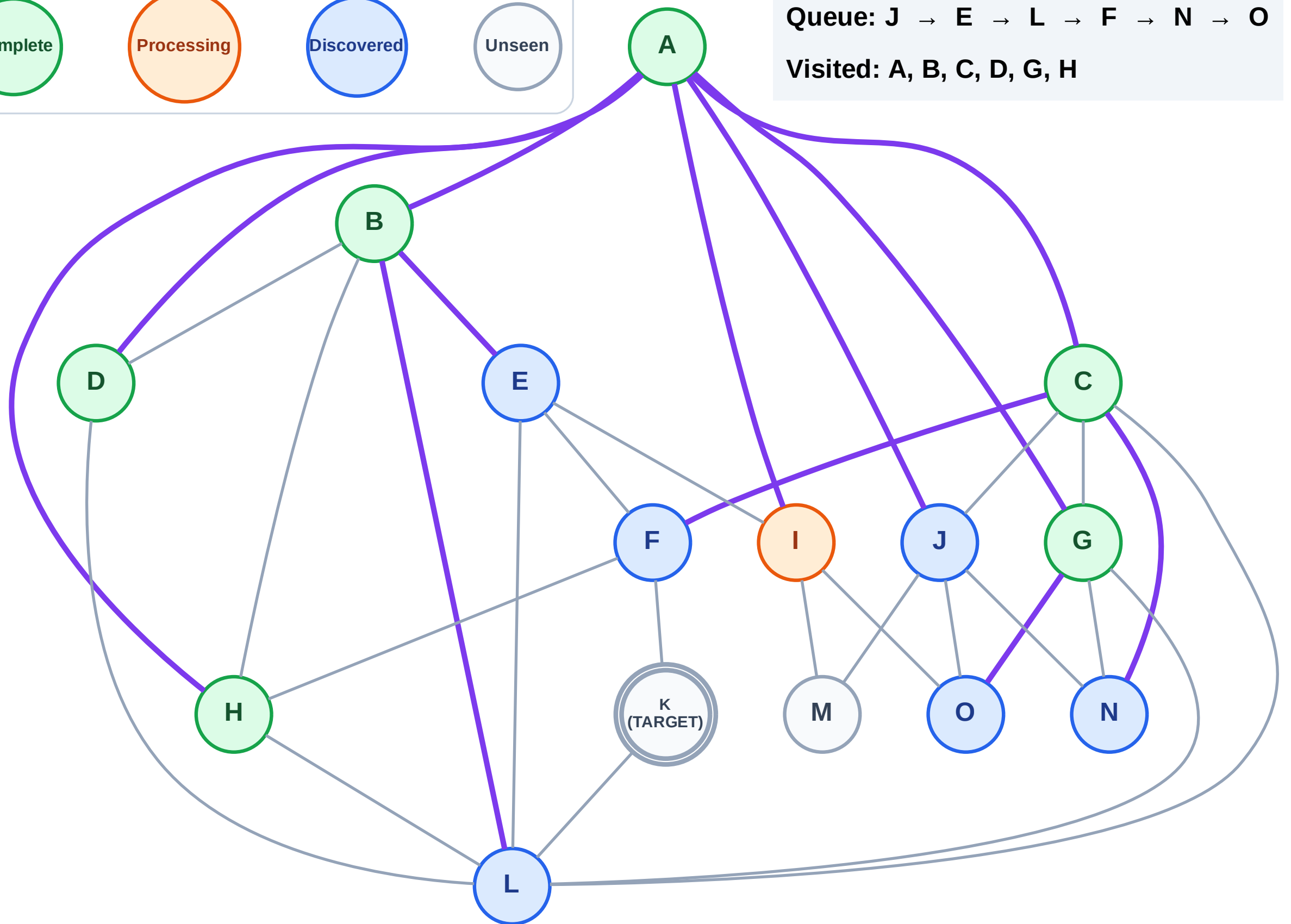
Step 14: Process node I

Legend



Queue: J → E → L → F → N → O

Visited: A, B, C, D, G, H



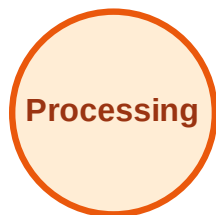
BFS Traversal

Step 15: Discover M from I

Legend



Complete



Processing



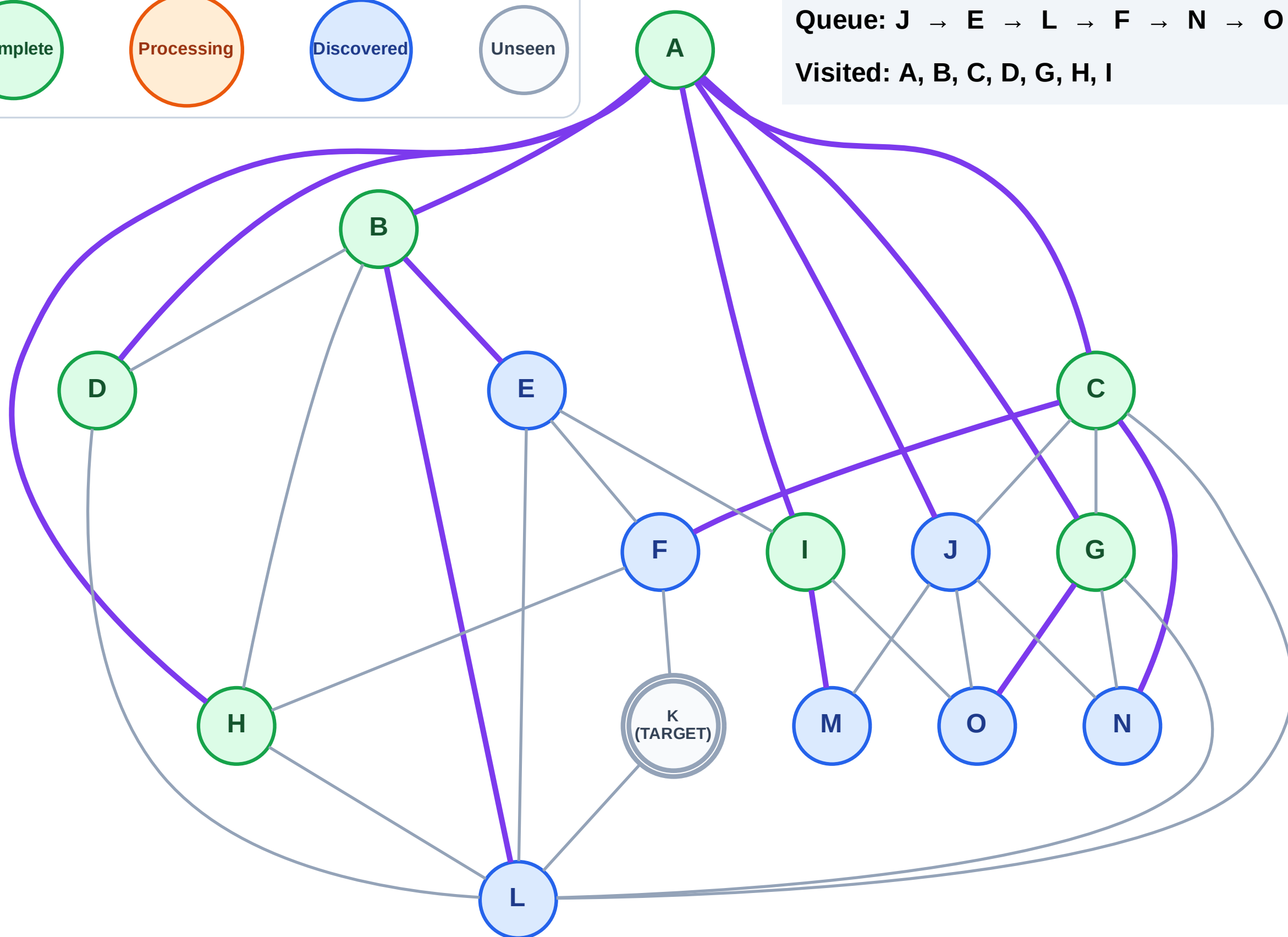
Discovered



Unseen

Queue: J → E → L → F → N → O → M

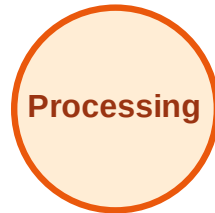
Visited: A, B, C, D, G, H, I



BFS Traversal

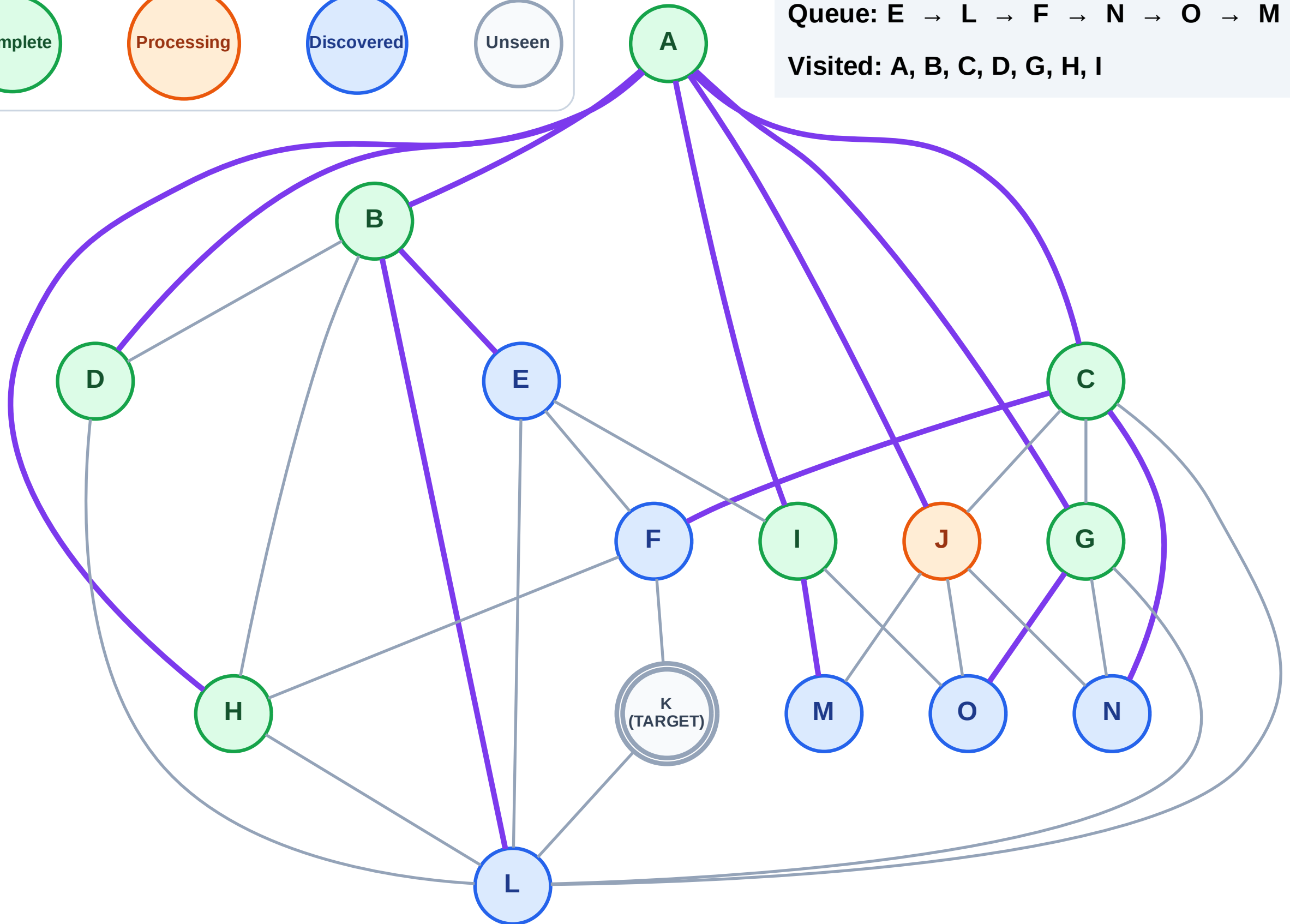
Step 16: Process node J

Legend



Queue: E → L → F → N → O → M

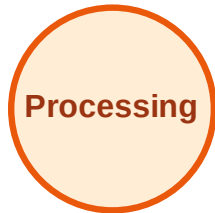
Visited: A, B, C, D, G, H, I



BFS Traversal

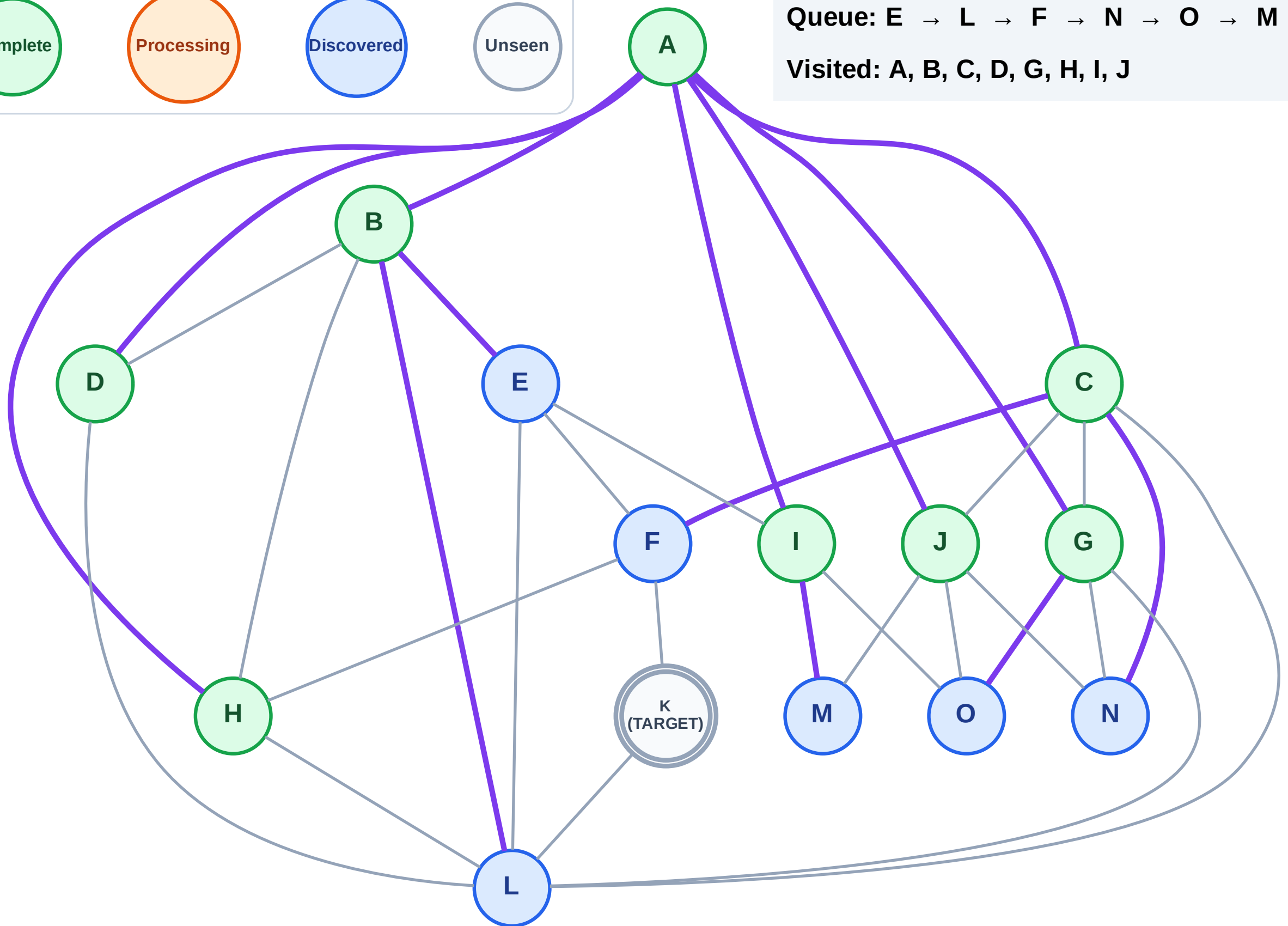
Step 17: No new neighbors from J

Legend



Queue: E → L → F → N → O → M

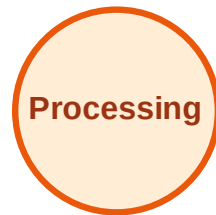
Visited: A, B, C, D, G, H, I, J



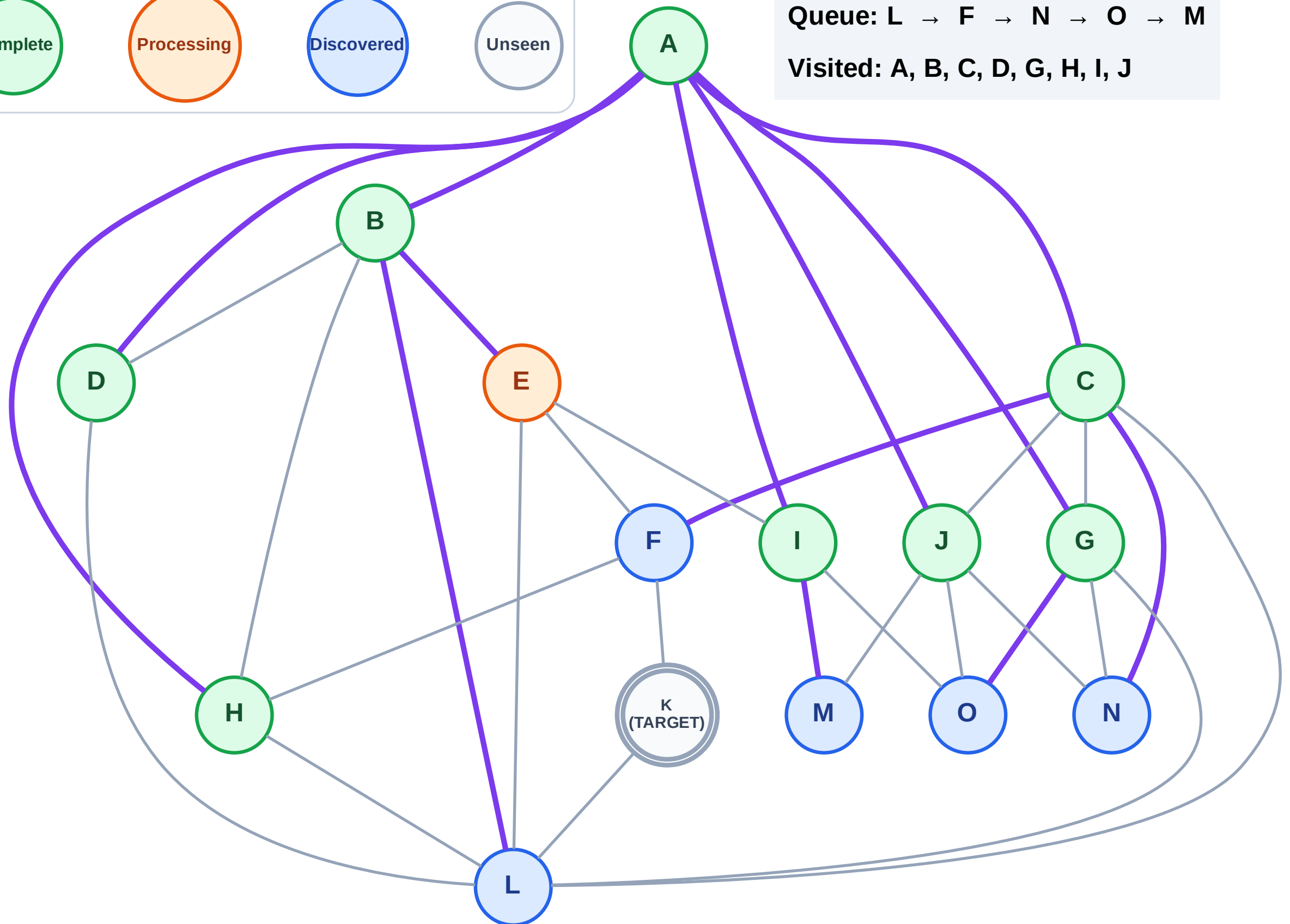
BFS Traversal

Step 18: Process node E

Legend



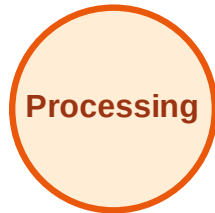
Queue: L → F → N → O → M
Visited: A, B, C, D, G, H, I, J



BFS Traversal

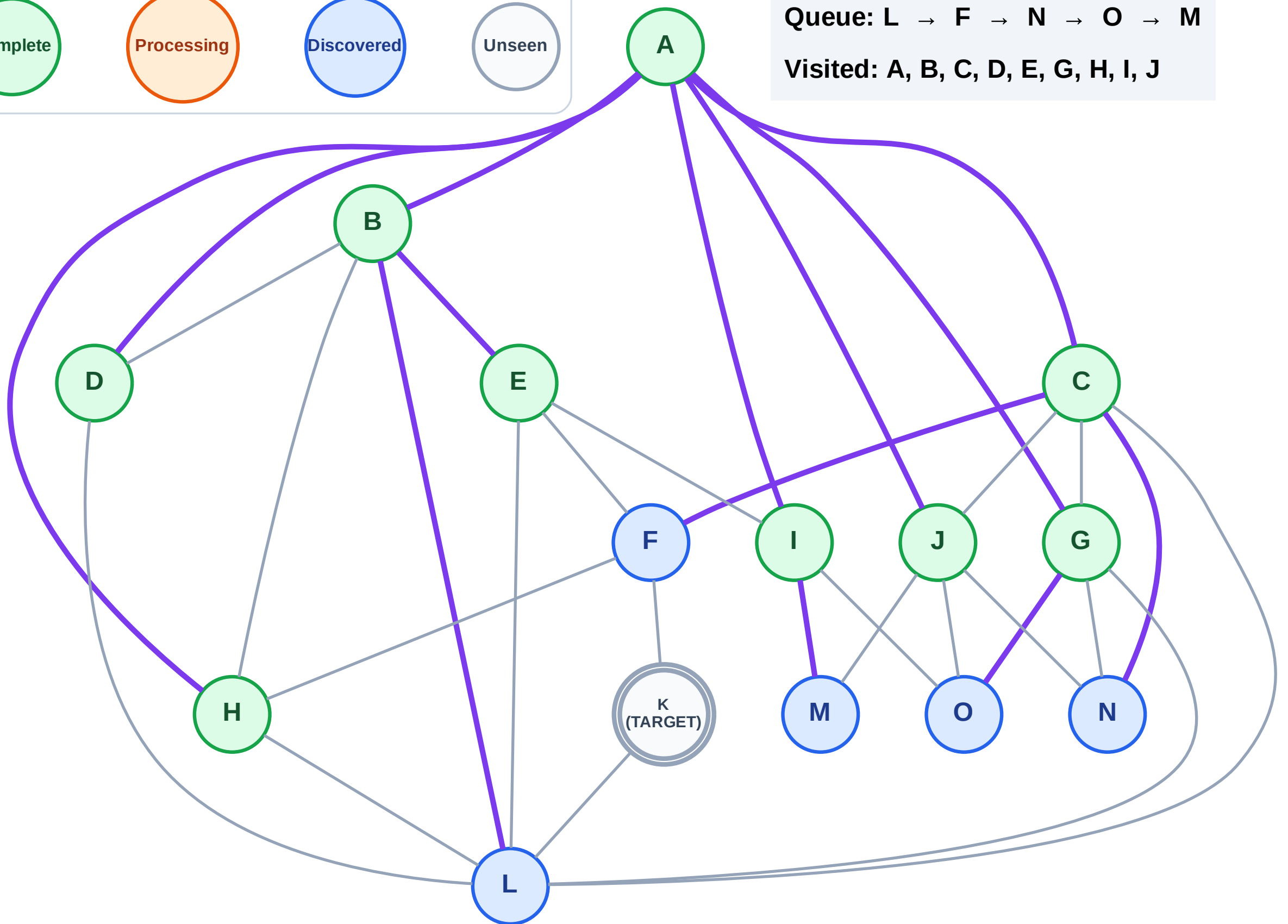
Step 19: No new neighbors from E

Legend



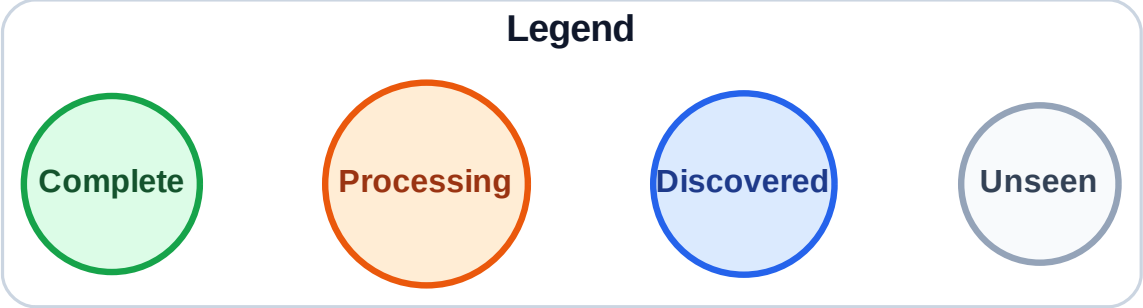
Queue: L → F → N → O → M

Visited: A, B, C, D, E, G, H, I, J

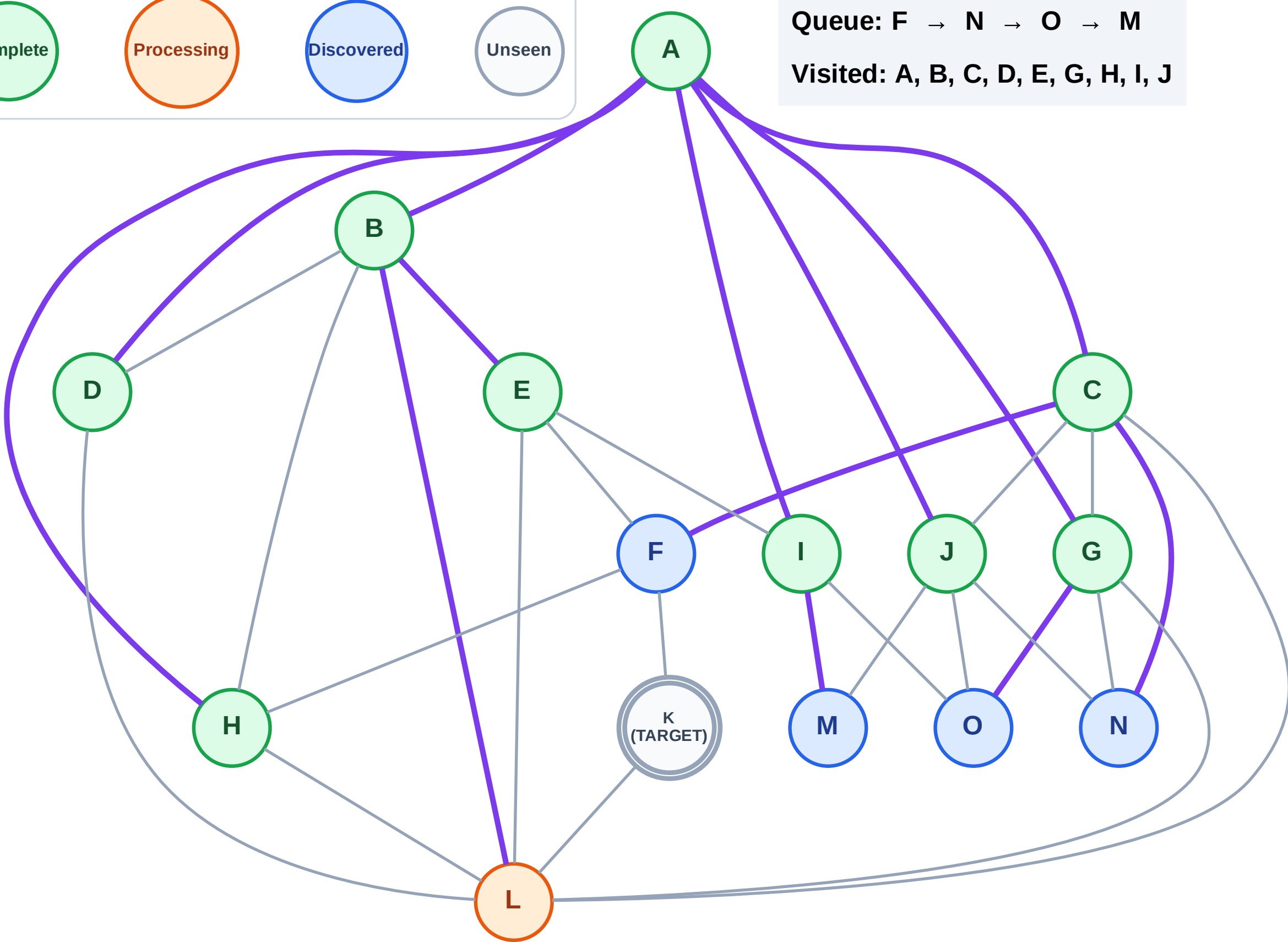


BFS Traversal

Step 20: Process node L



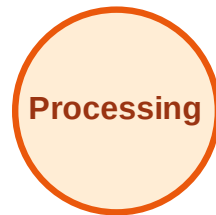
Queue: F → N → O → M
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

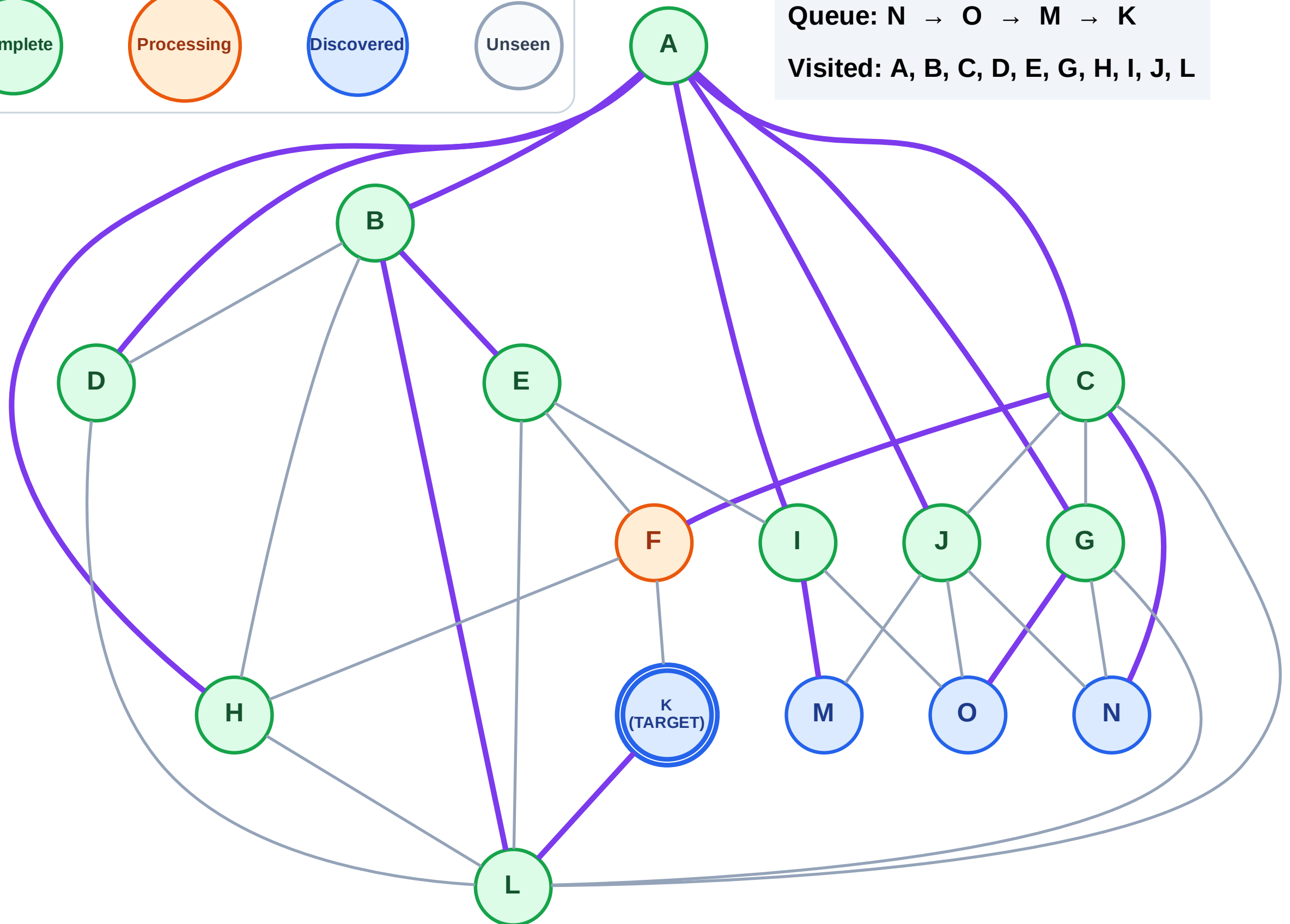
Step 22: Process node F

Legend



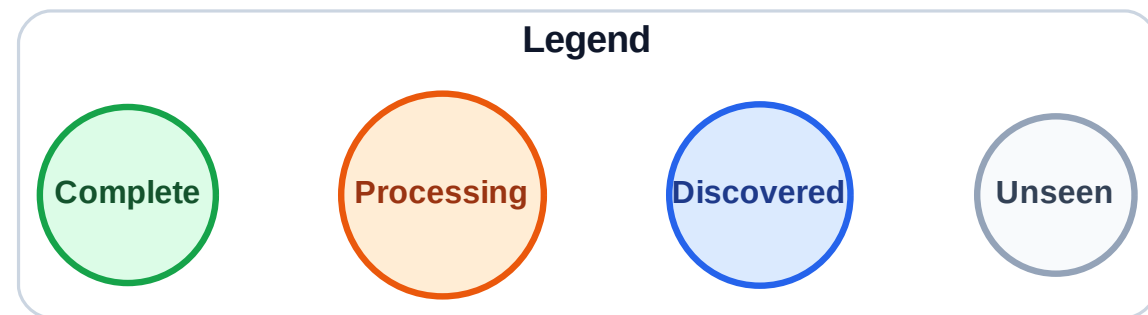
Queue: N → O → M → K

Visited: A, B, C, D, E, G, H, I, J, L

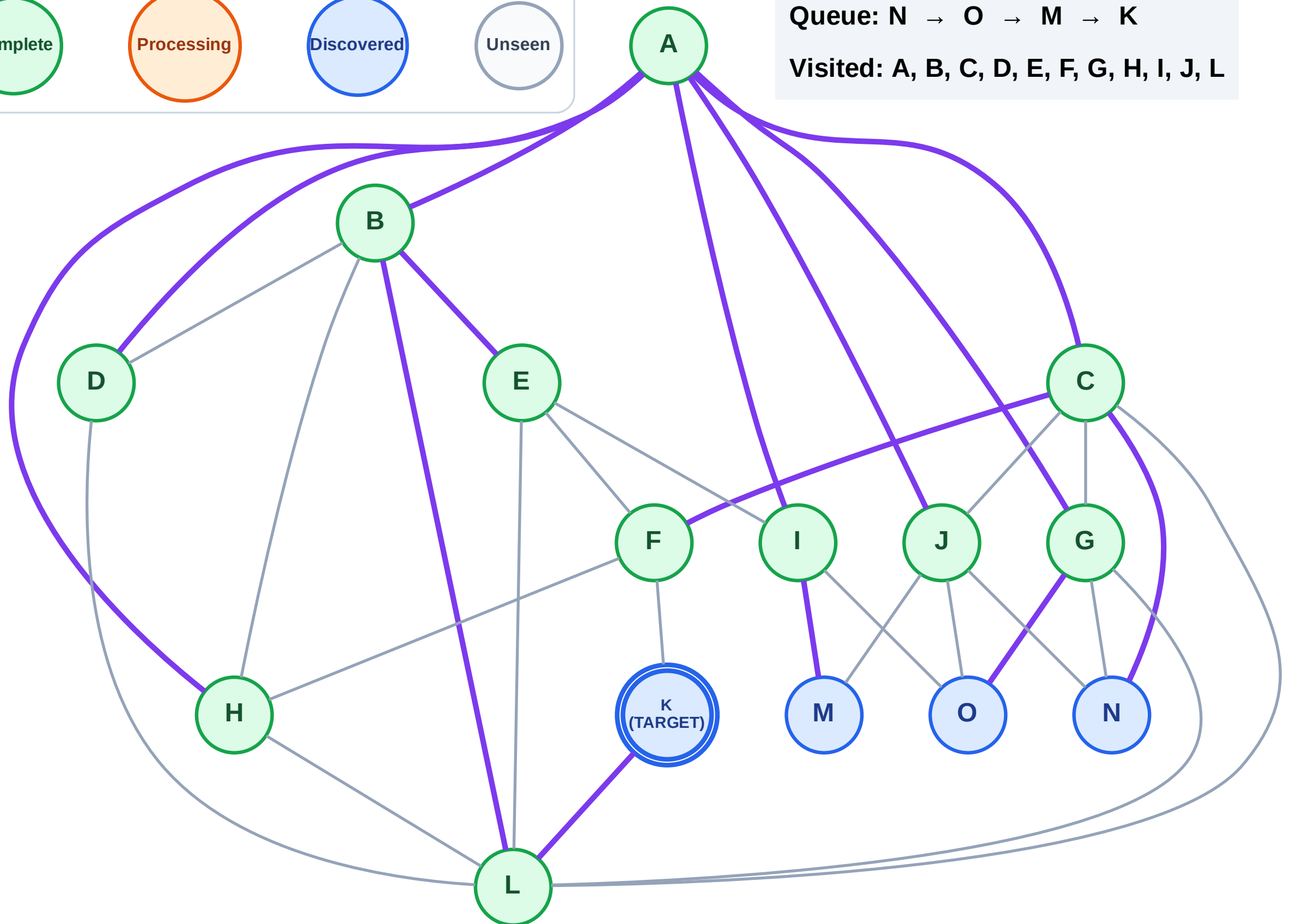


BFS Traversal

Step 23: No new neighbors from F



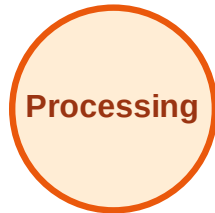
Queue: N → O → M → K
Visited: A, B, C, D, E, F, G, H, I, J, L



BFS Traversal

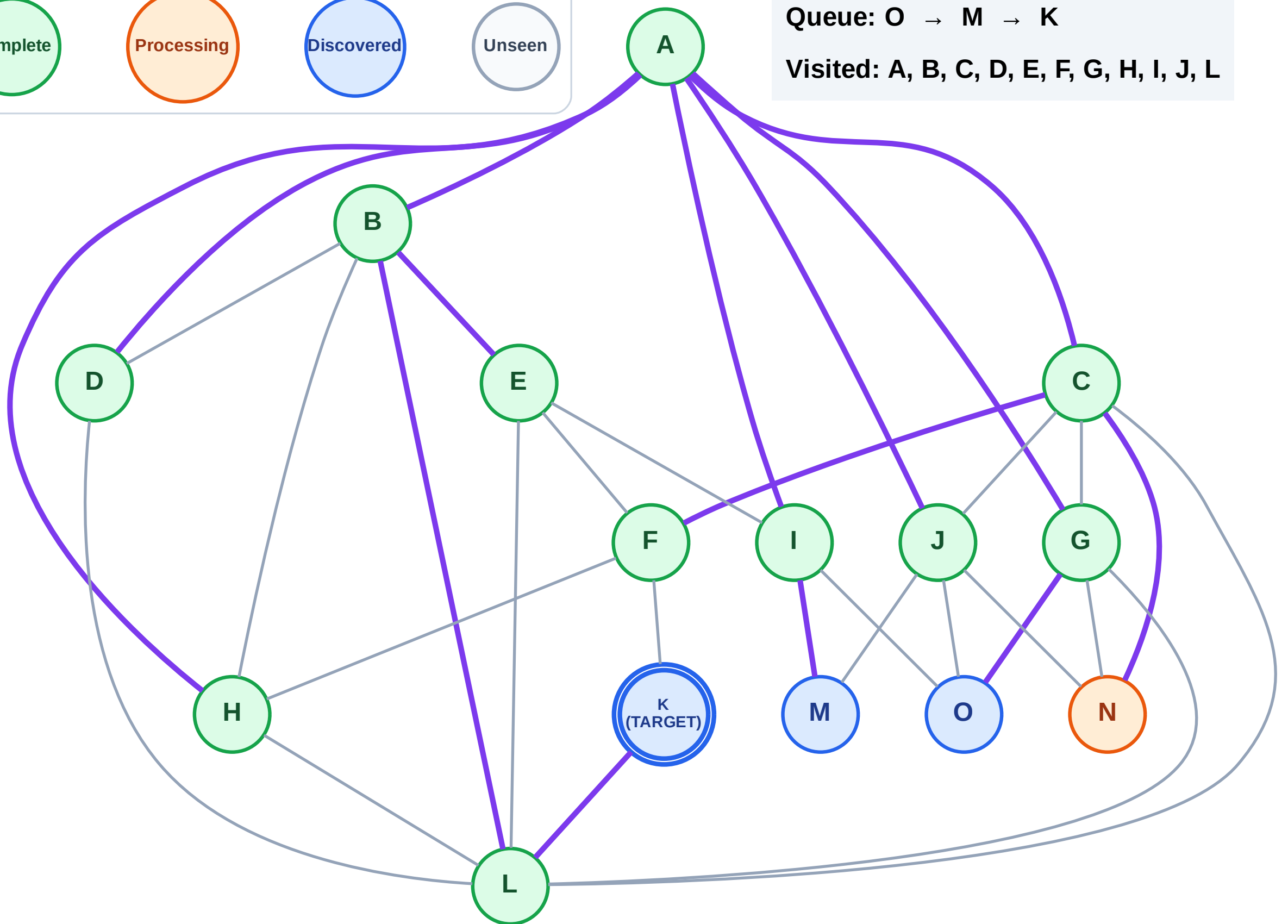
Step 24: Process node N

Legend



Queue: O → M → K

Visited: A, B, C, D, E, F, G, H, I, J, L

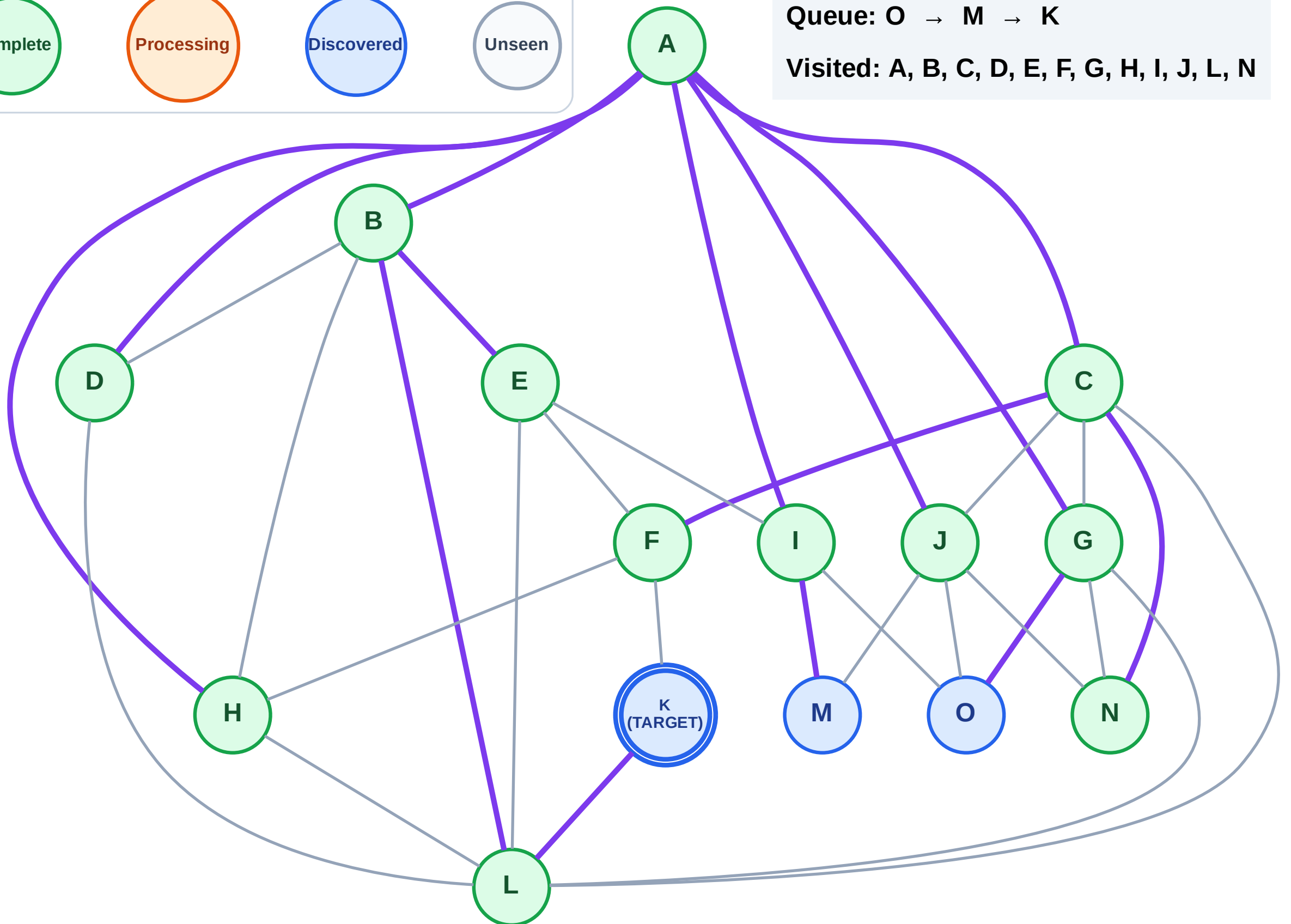


Step 25: No new neighbors from N

Legend

Unseen

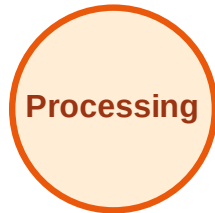
Visited: A, B, C, D, E, F, G, H, I, J, L, N



BFS Traversal

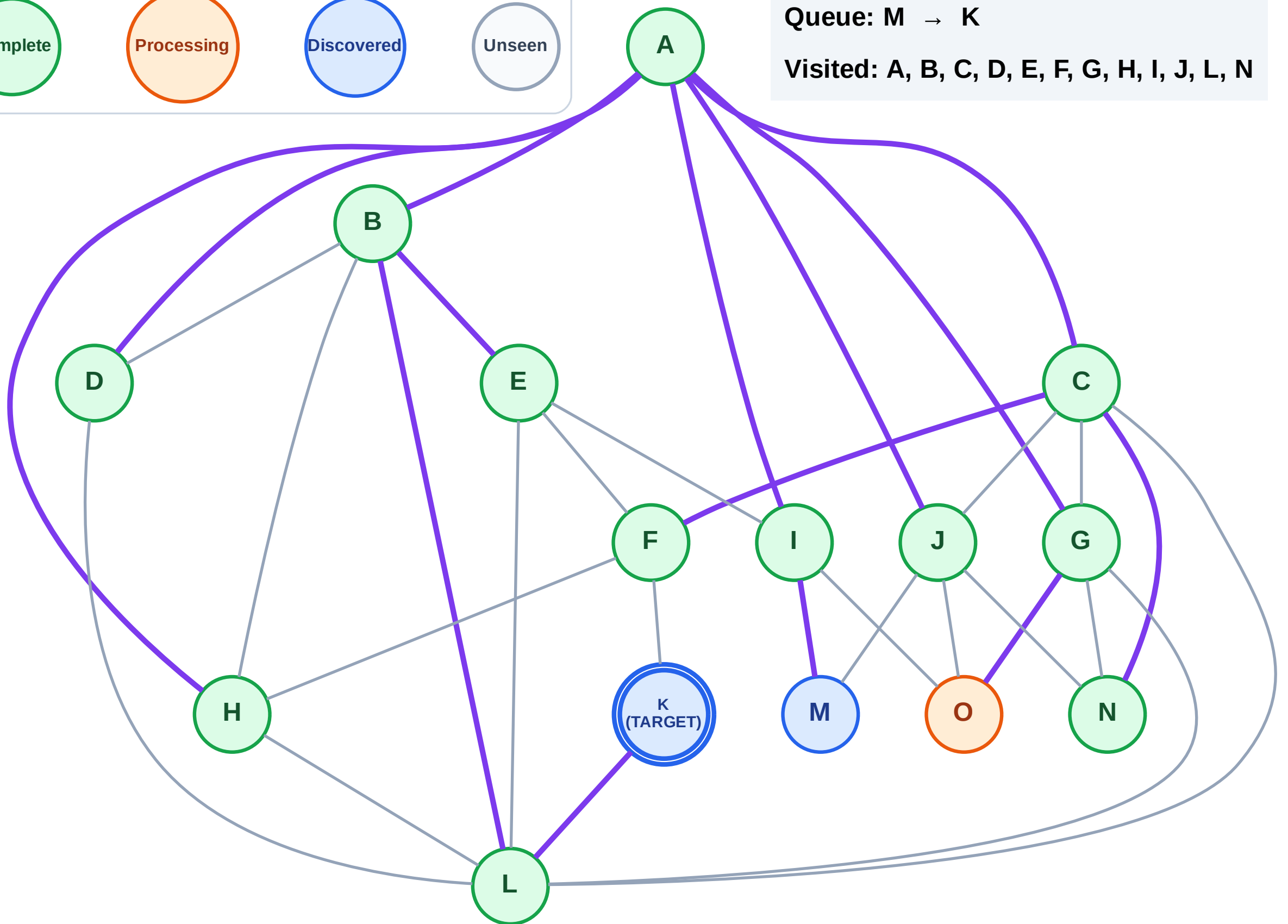
Step 26: Process node O

Legend



Queue: M → K

Visited: A, B, C, D, E, F, G, H, I, J, L, N



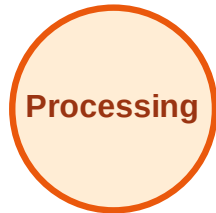
BFS Traversal

Step 27: No new neighbors from O

Legend



Complete



Processing



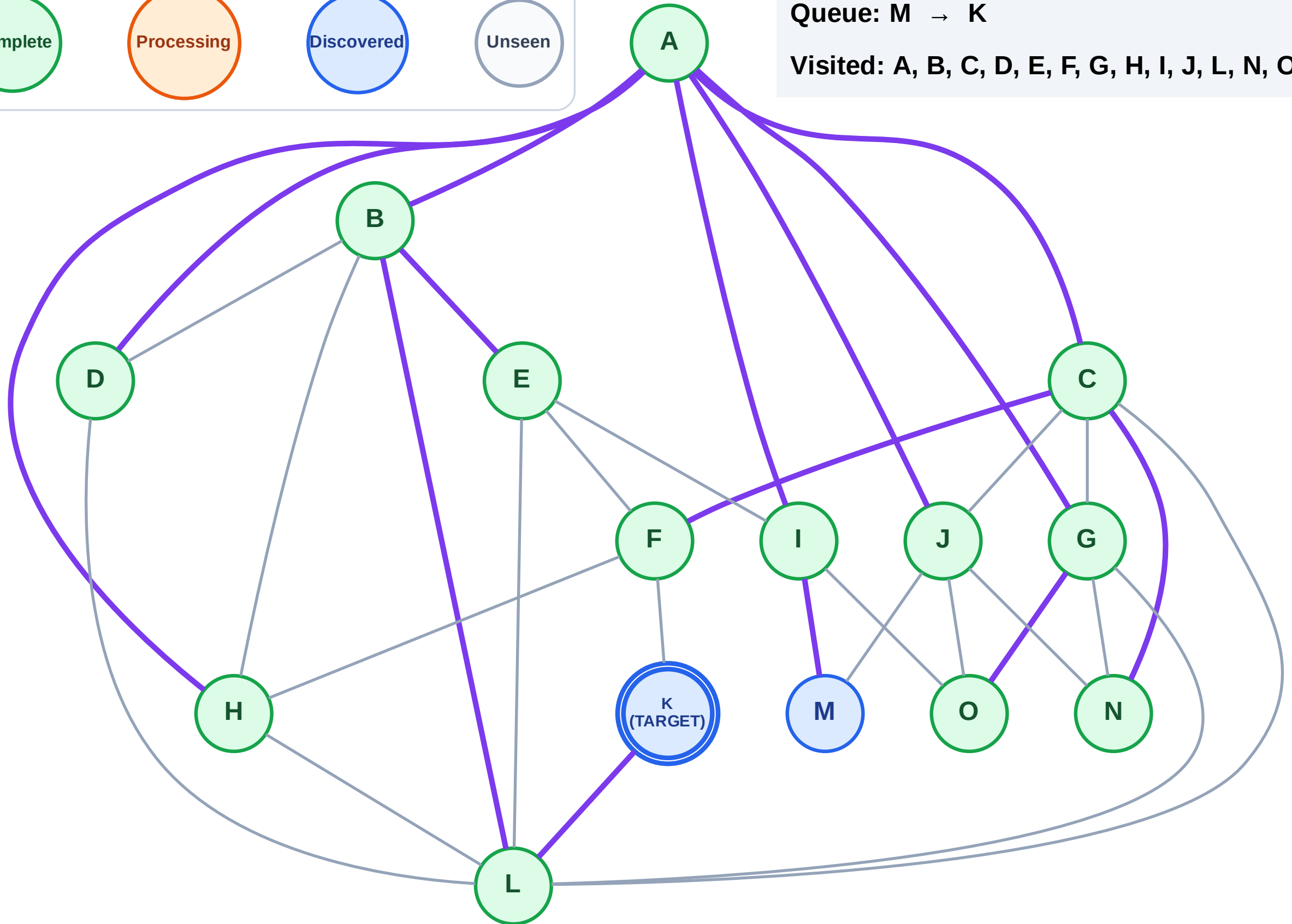
Discovered



Unseen

Queue: M → K

Visited: A, B, C, D, E, F, G, H, I, J, L, N, O



BFS Traversal

Step 28: Process node M

Legend

Complete

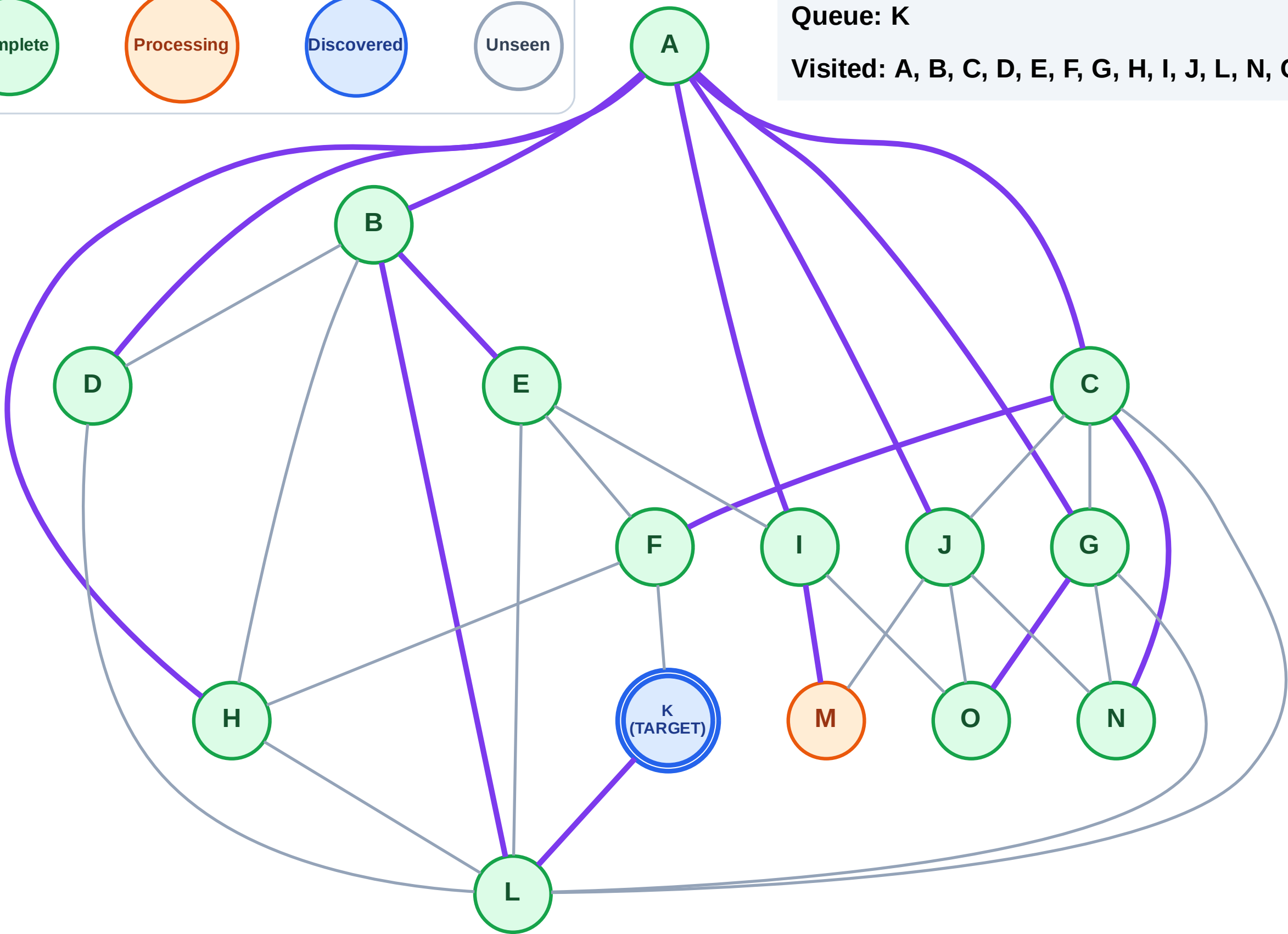
Processing

Discovered

Unseen

Queue: K

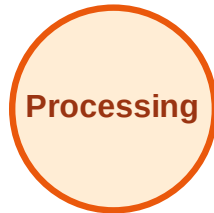
Visited: A, B, C, D, E, F, G, H, I, J, L, N, O



BFS Traversal

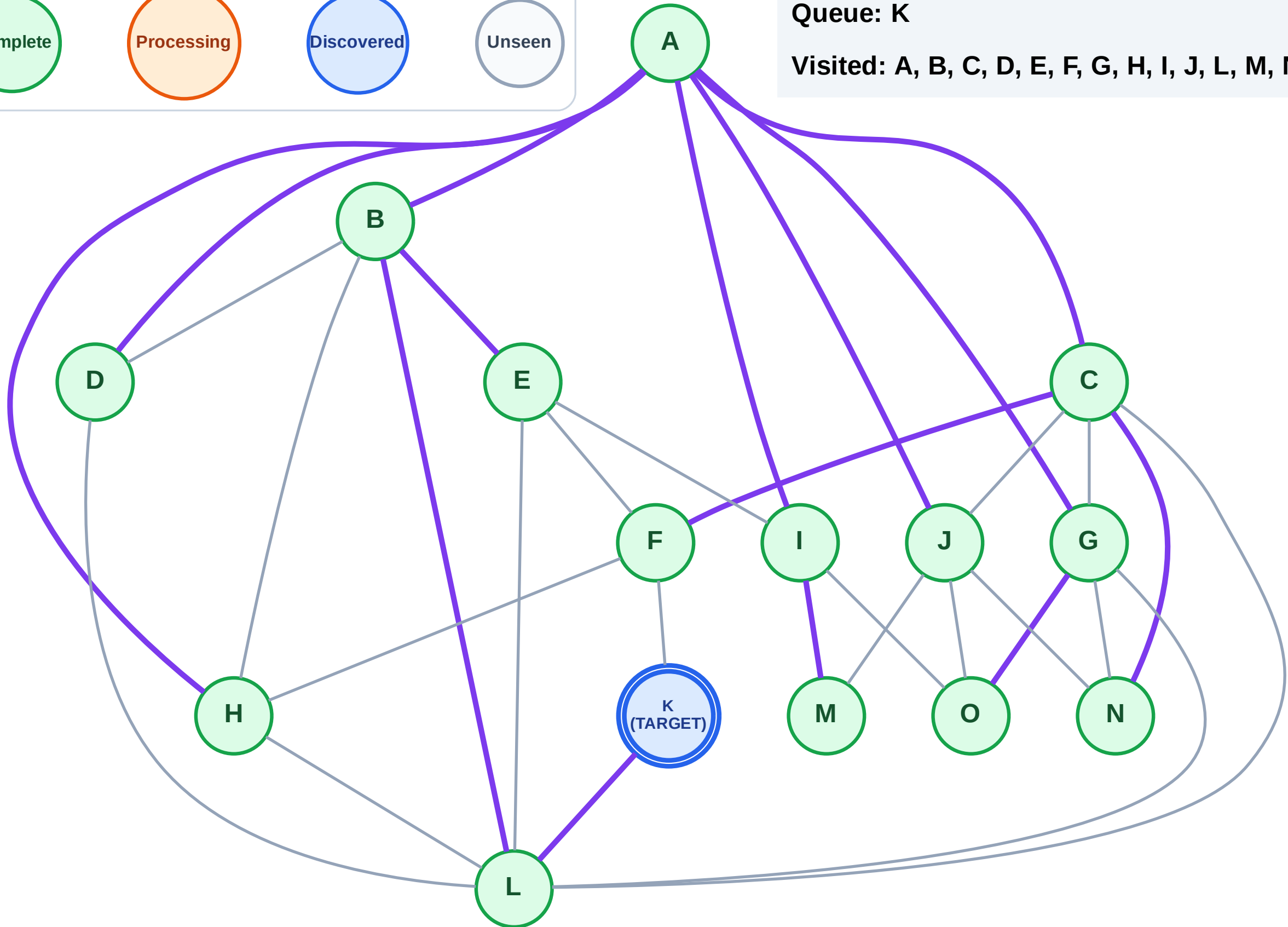
Step 29: No new neighbors from M

Legend



Queue: K

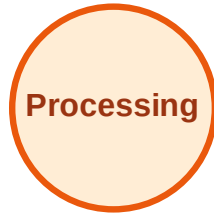
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

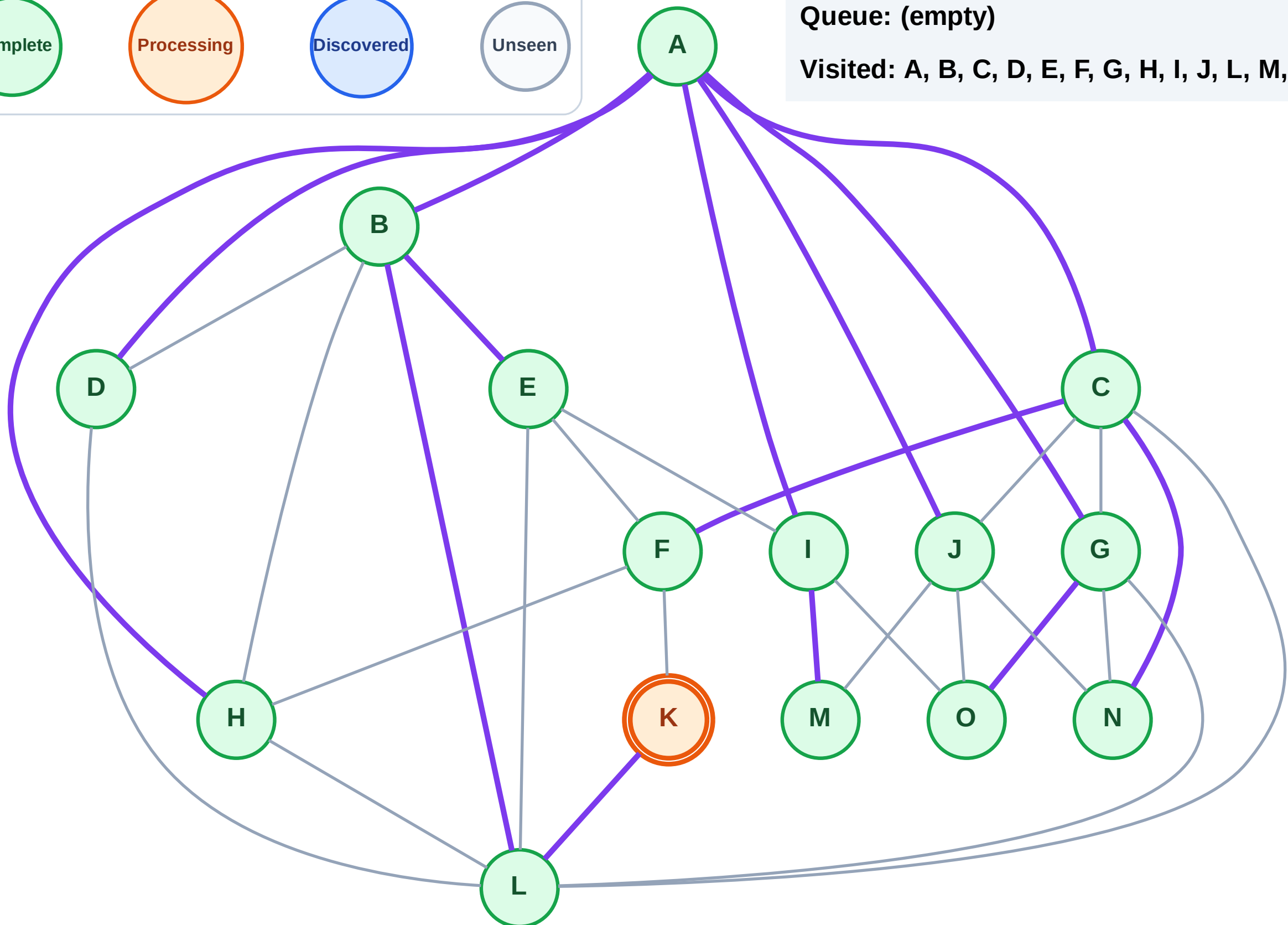
Step 30: Process node K

Legend



Queue: (empty)

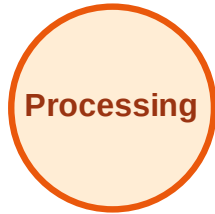
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

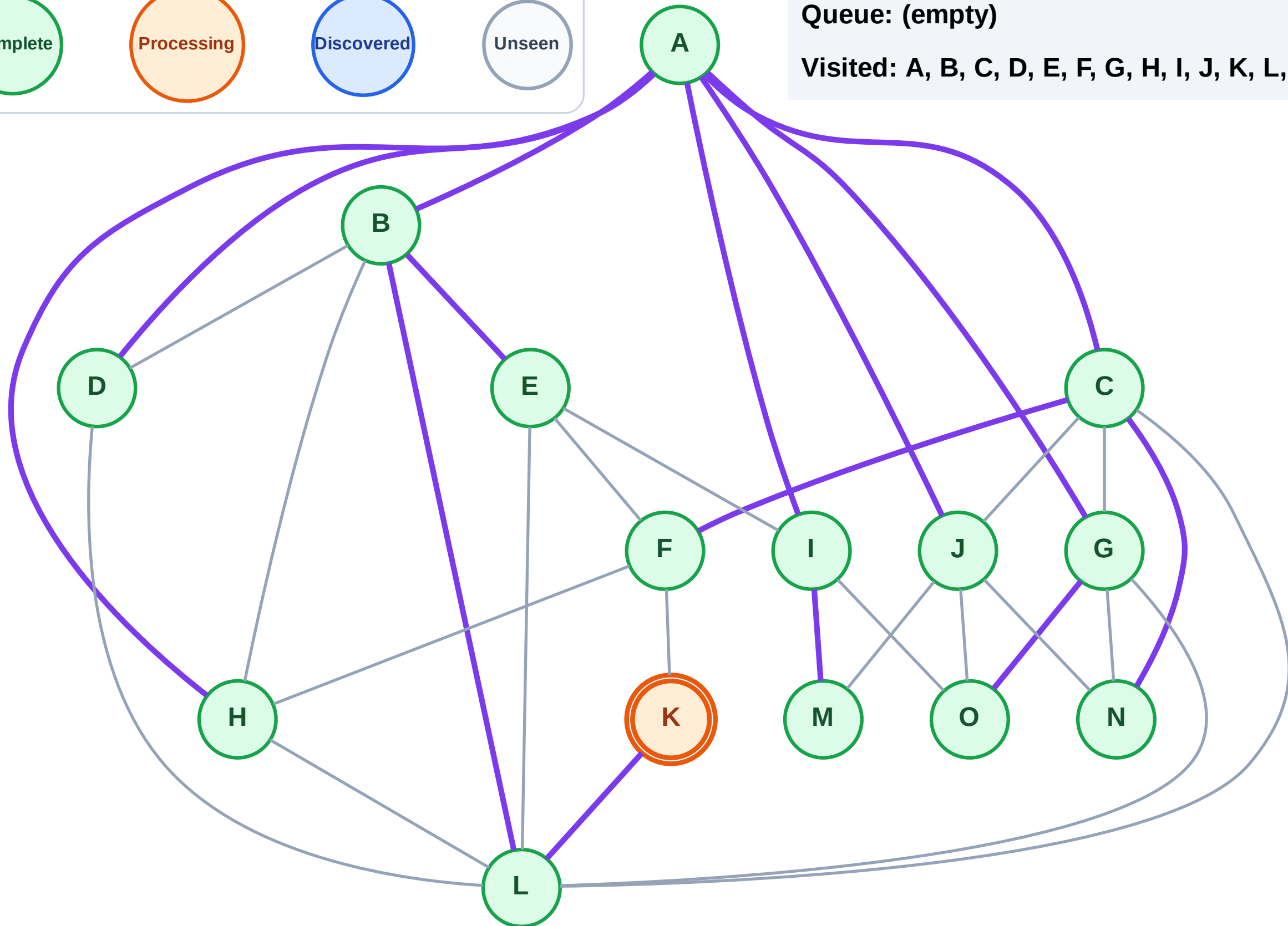
Step 31: TARGET NODE K REACHED! Path (3 hops): A → B → L → K

Legend



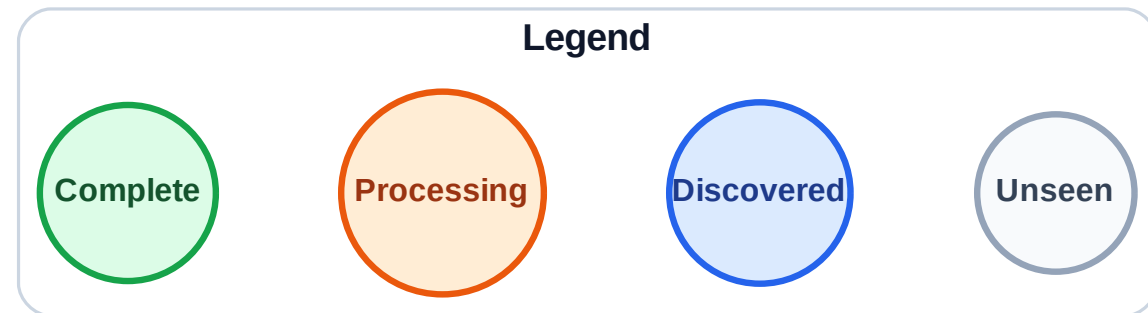
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

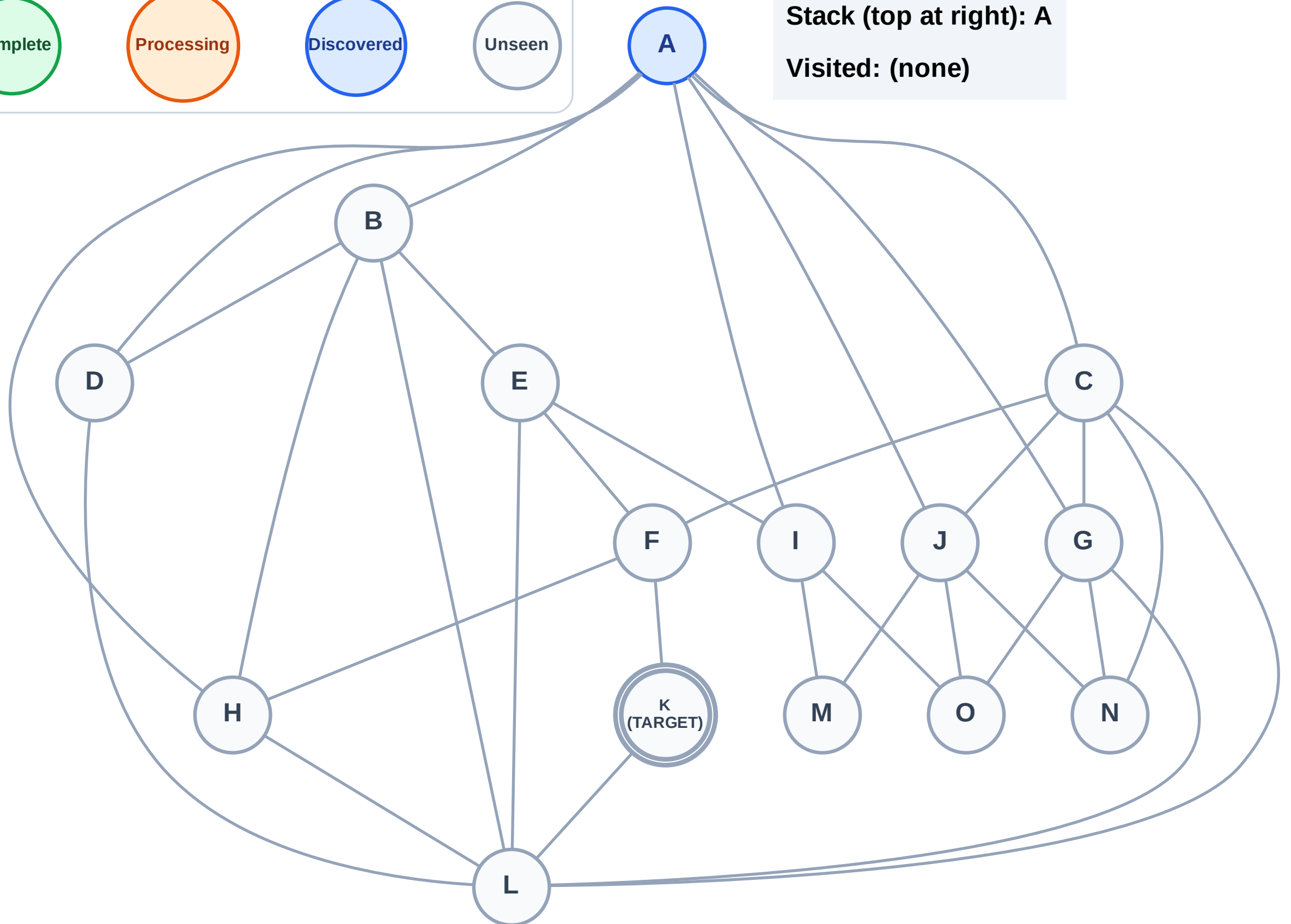


DFS Traversal

Step 1: Start at A (Seeking target node K)



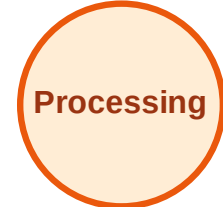
Stack (top at right): A
Visited: (none)



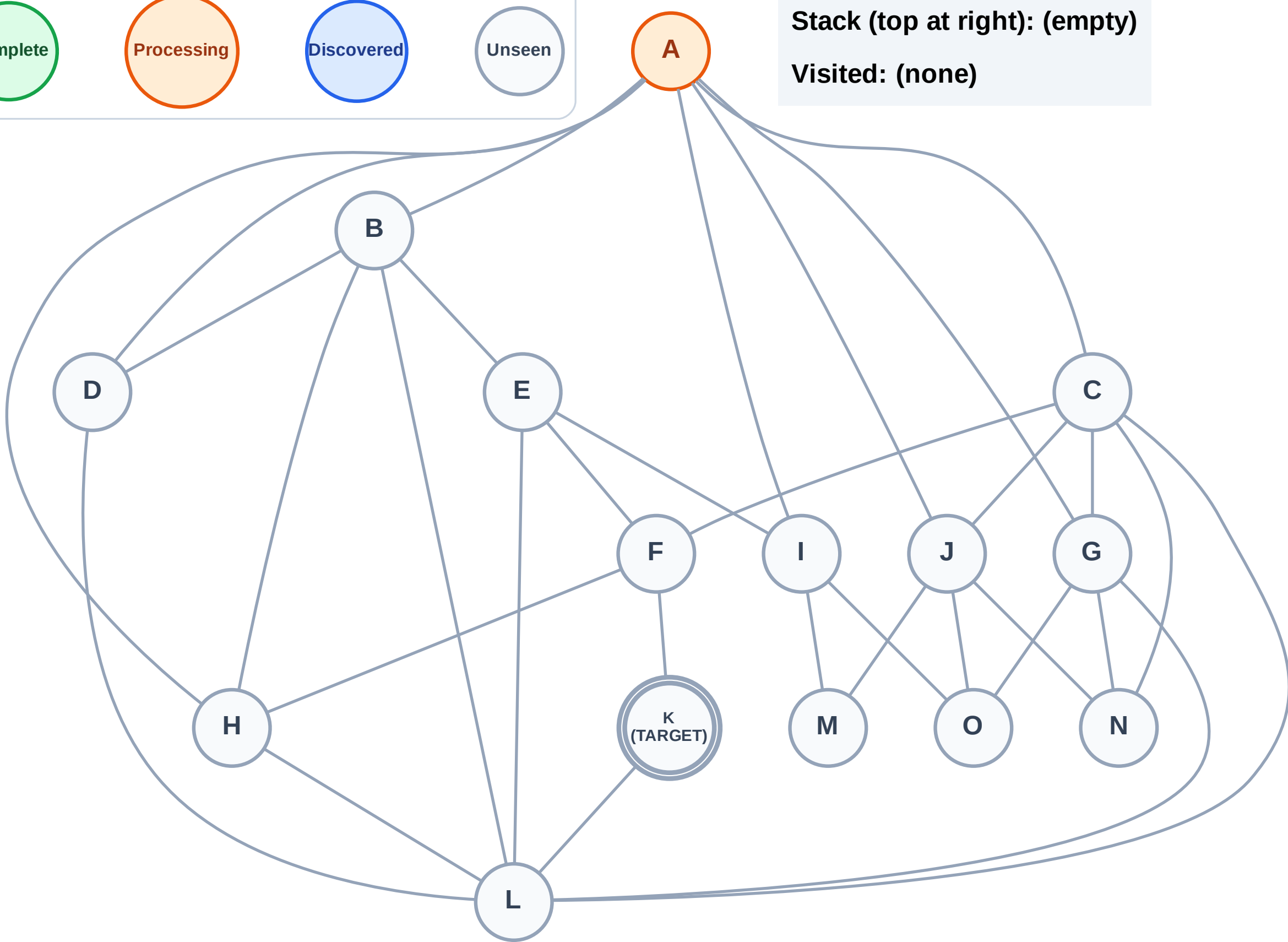
DFS Traversal

Step 2: Process node A

Legend



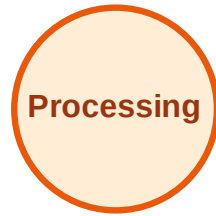
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

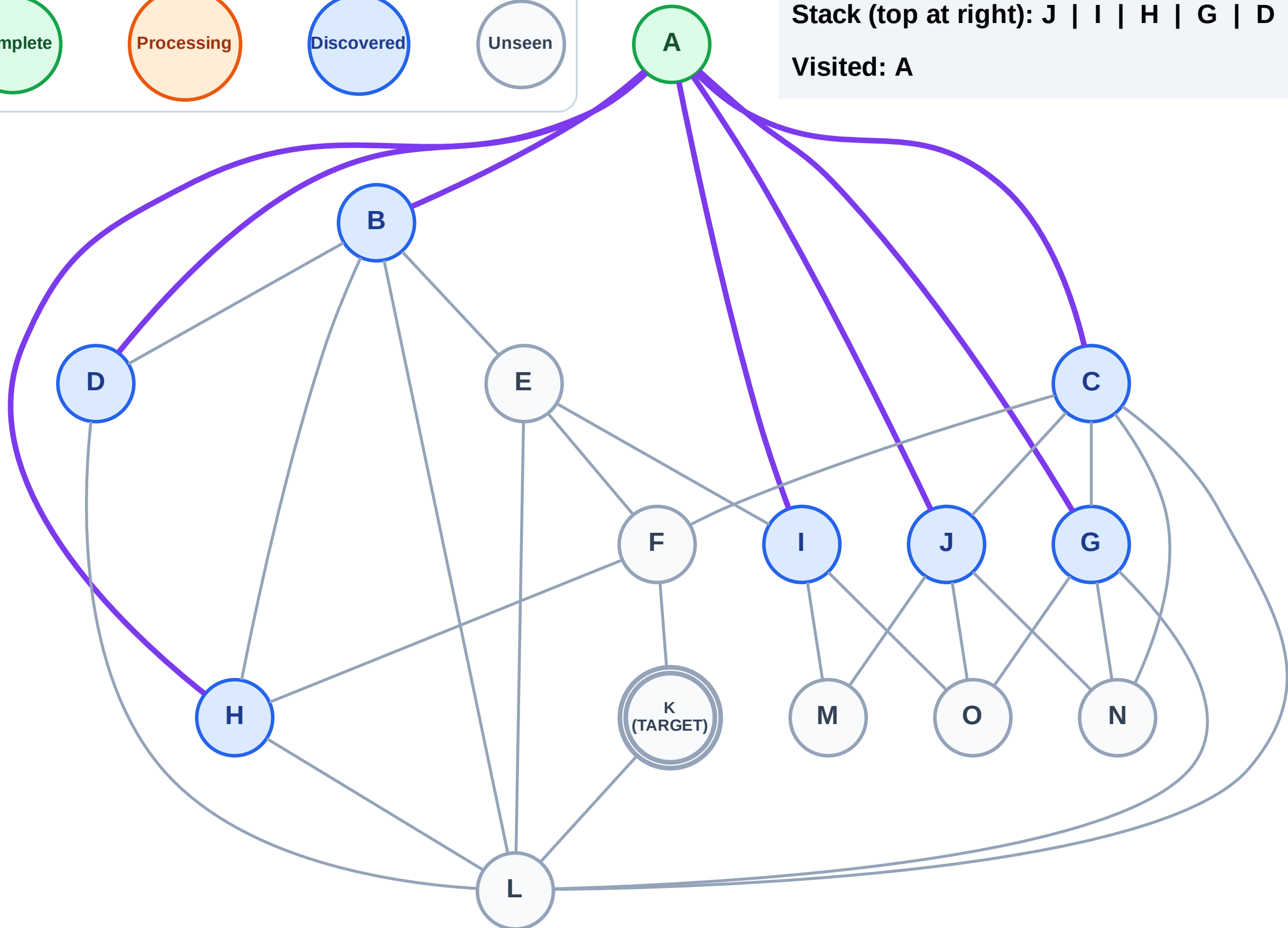
Step 3: Discover J, I, H, G, D, C, B from A

Legend



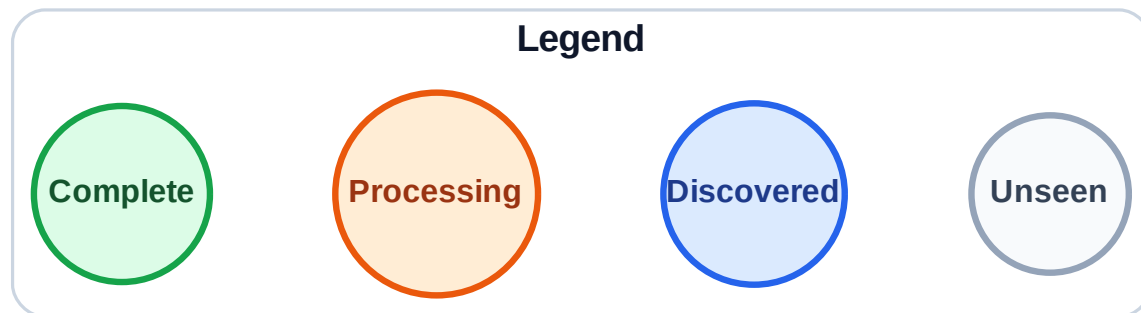
Stack (top at right): J | I | H | G | D | C | B

Visited: A



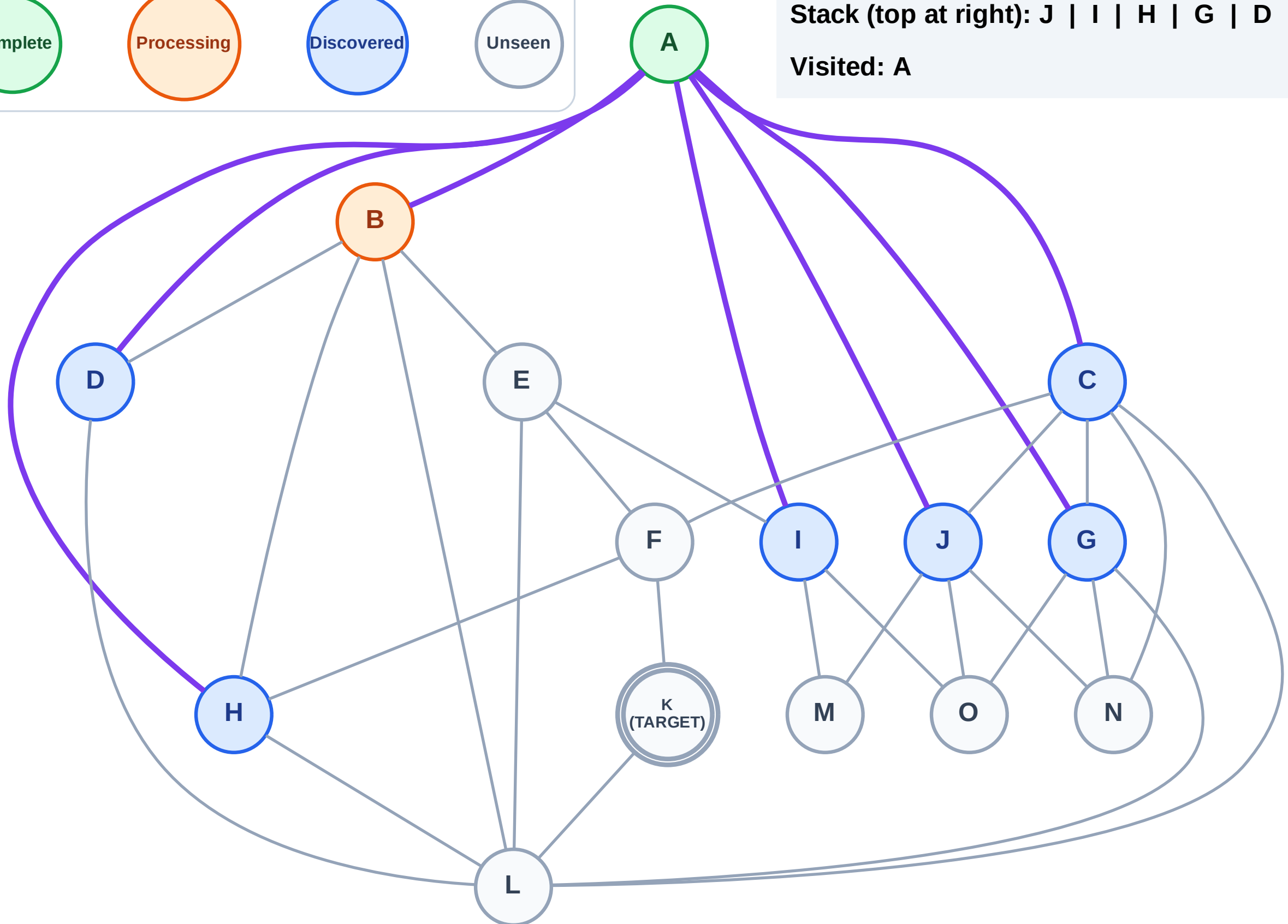
DFS Traversal

Step 4: Process node B



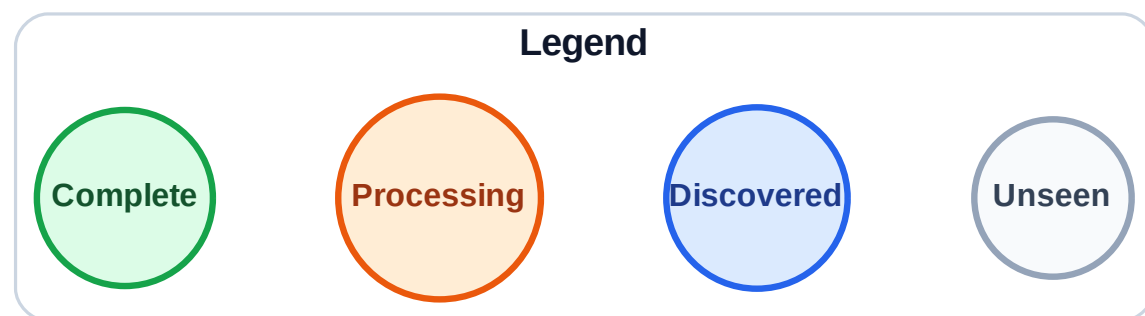
Stack (top at right): J | I | H | G | D | C

Visited: A



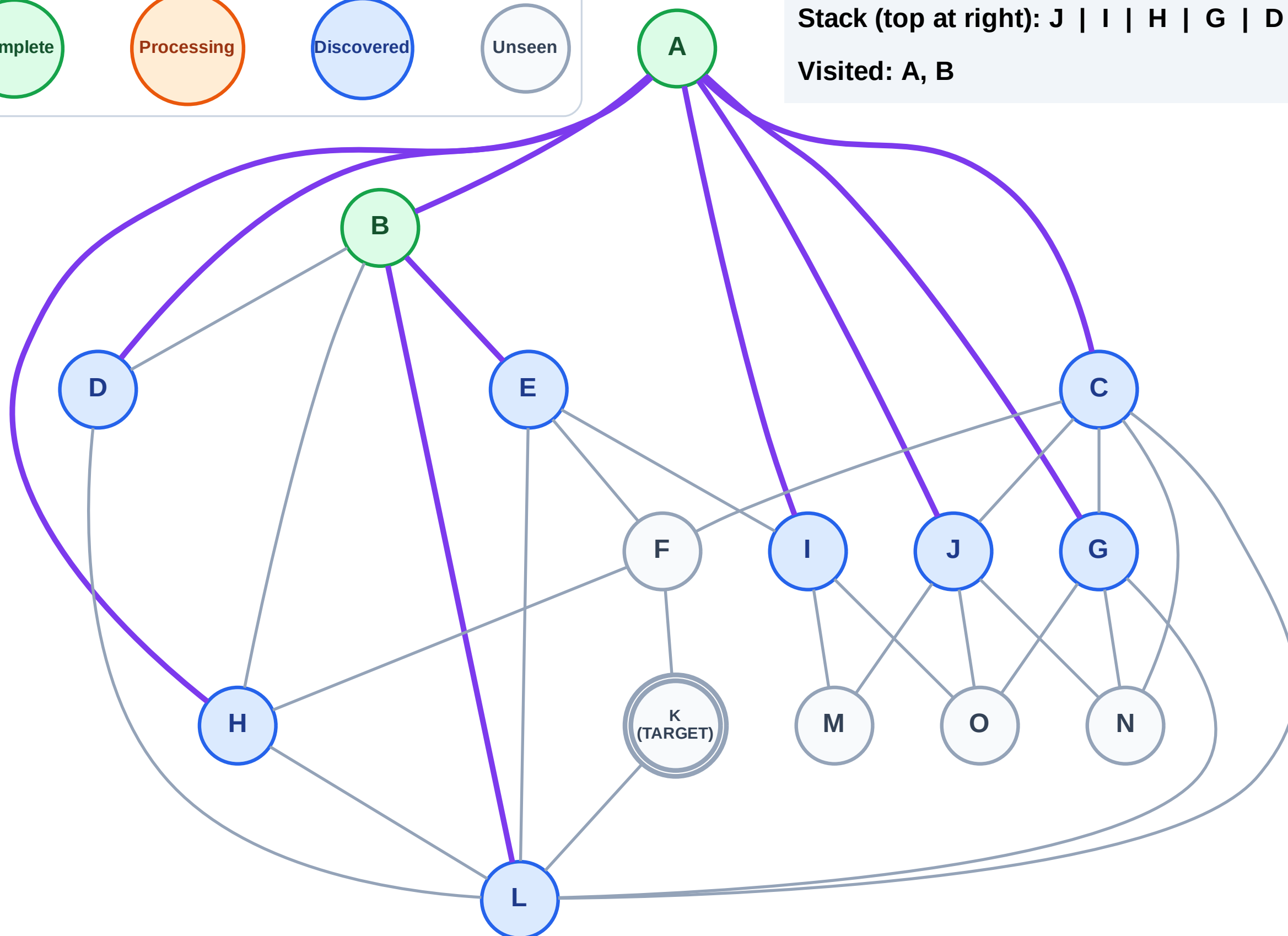
DFS Traversal

Step 5: Discover L, E from B



Stack (top at right): J | I | H | G | D | C | L | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

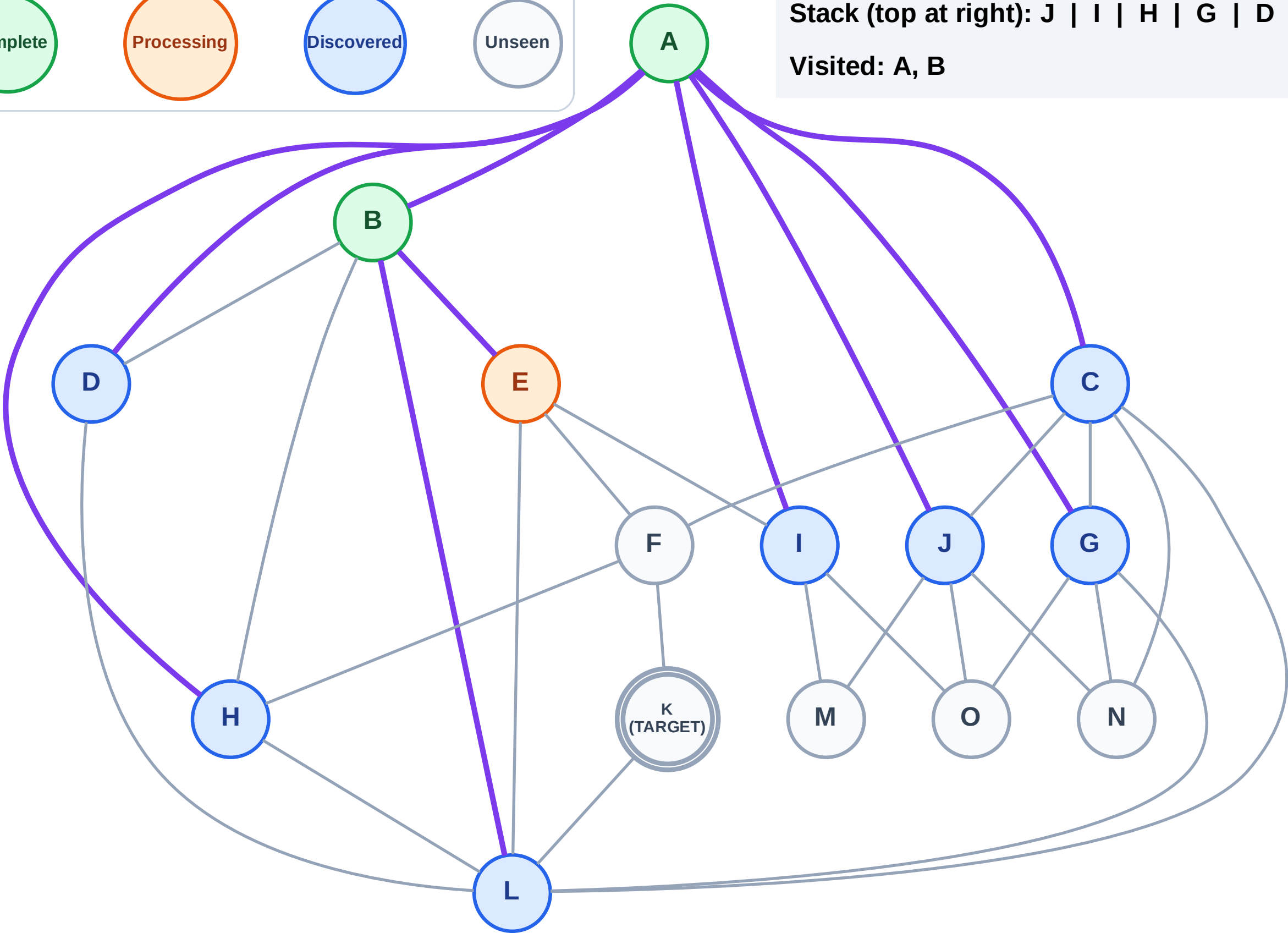
Processing

Discovered

Unseen

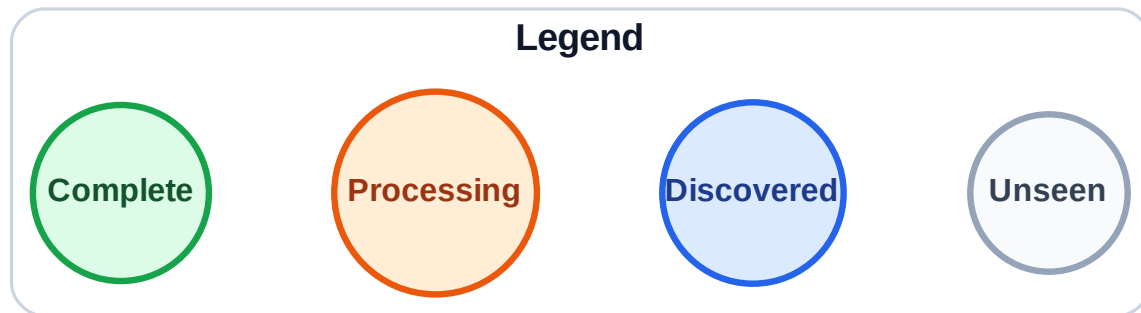
Stack (top at right): J | I | H | G | D | C | L

Visited: A, B



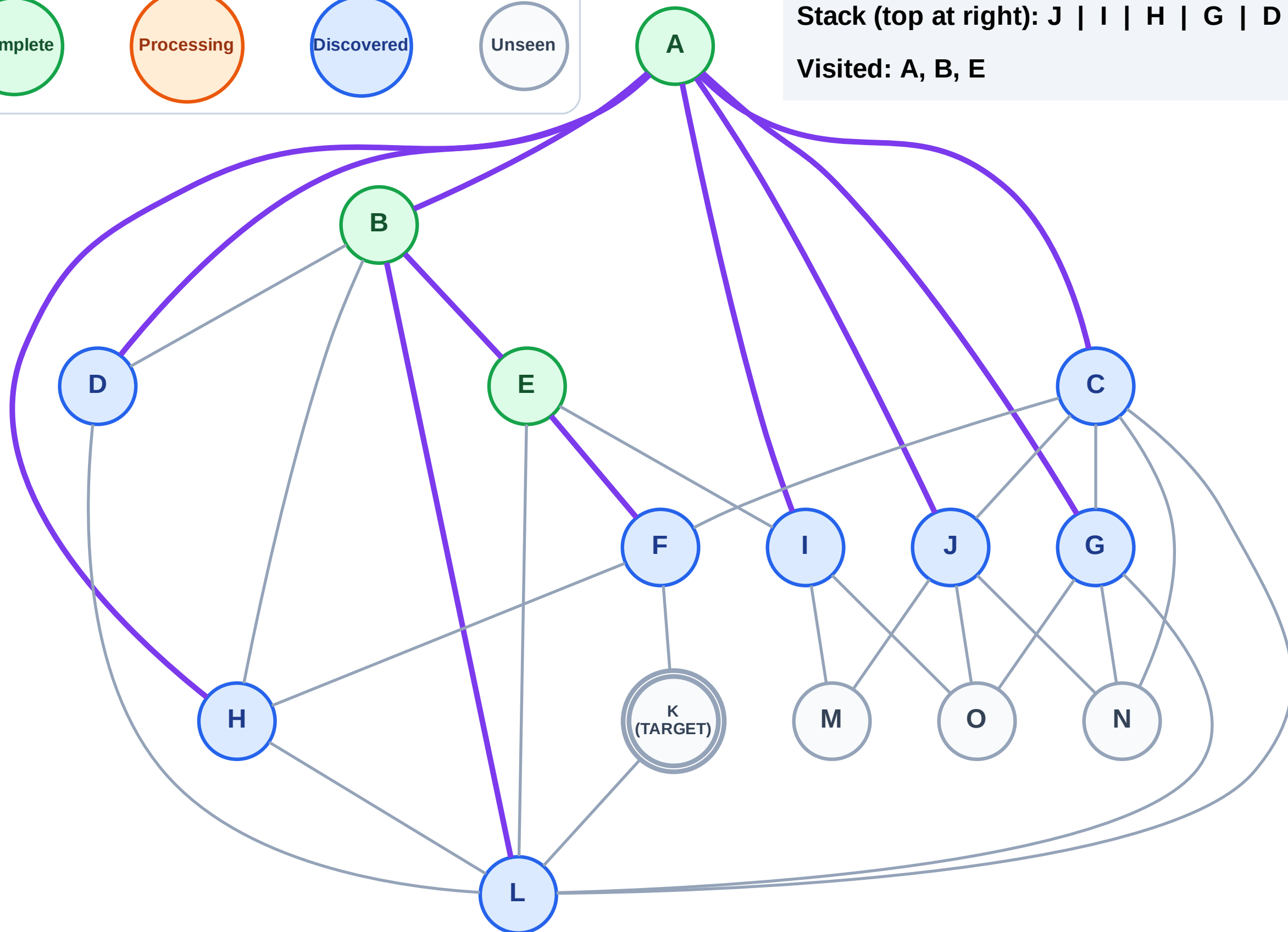
DFS Traversal

Step 7: Discover F from E



Stack (top at right): J | I | H | G | D | C | L | F

Visited: A, B, E



DFS Traversal

Step 8: Process node F

Legend

Complete

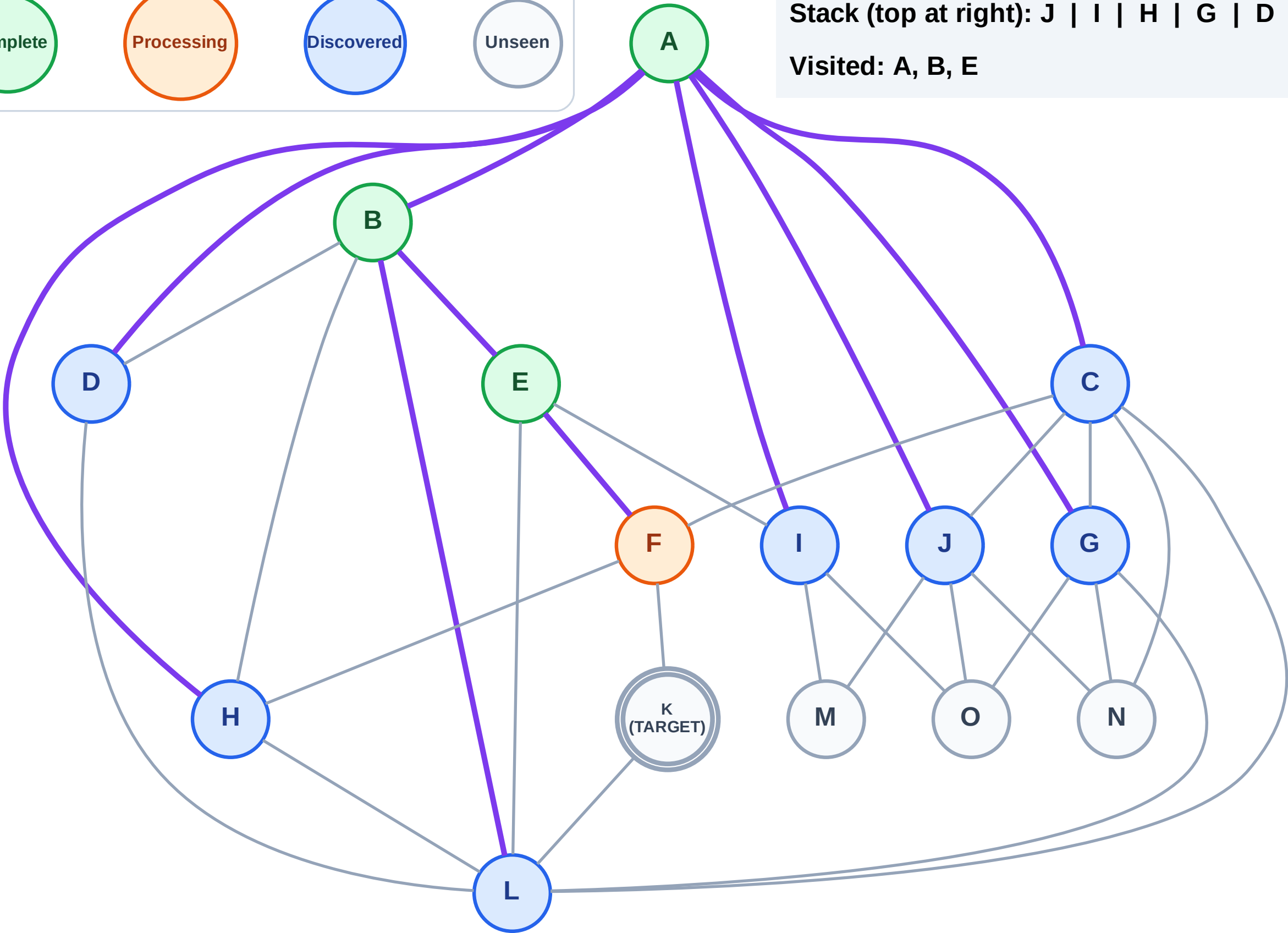
Processing

Discovered

Unseen

Stack (top at right): J | I | H | G | D | C | L

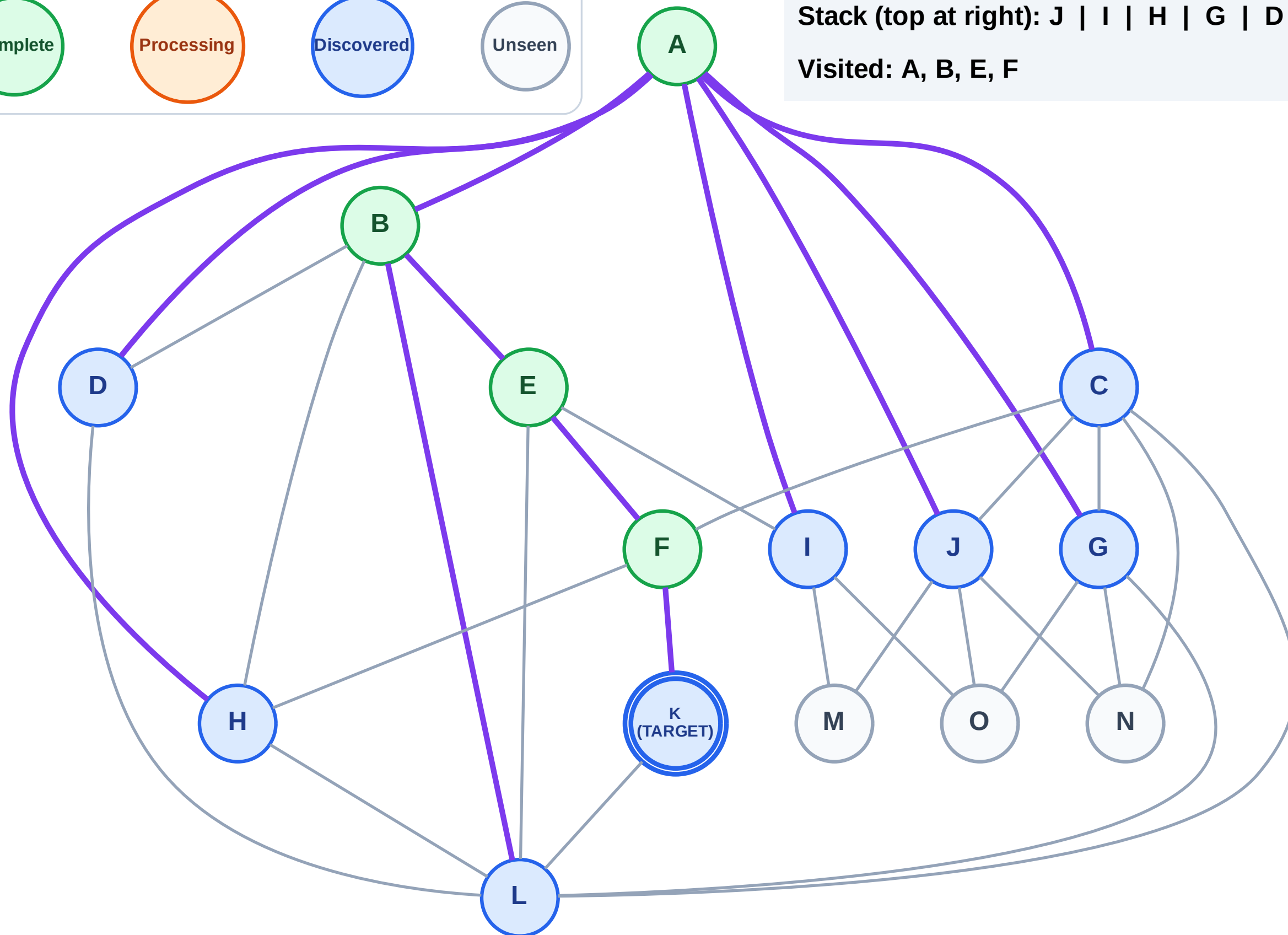
Visited: A, B, E



Step 9: Discover K from F

Stack (top at right): J | I | H | G | D | C | L | K

Visited: A, B, E, F



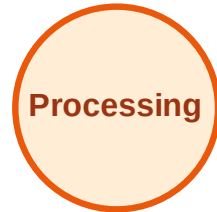
DFS Traversal

Step 10: Process node K

Legend



Complete



Processing



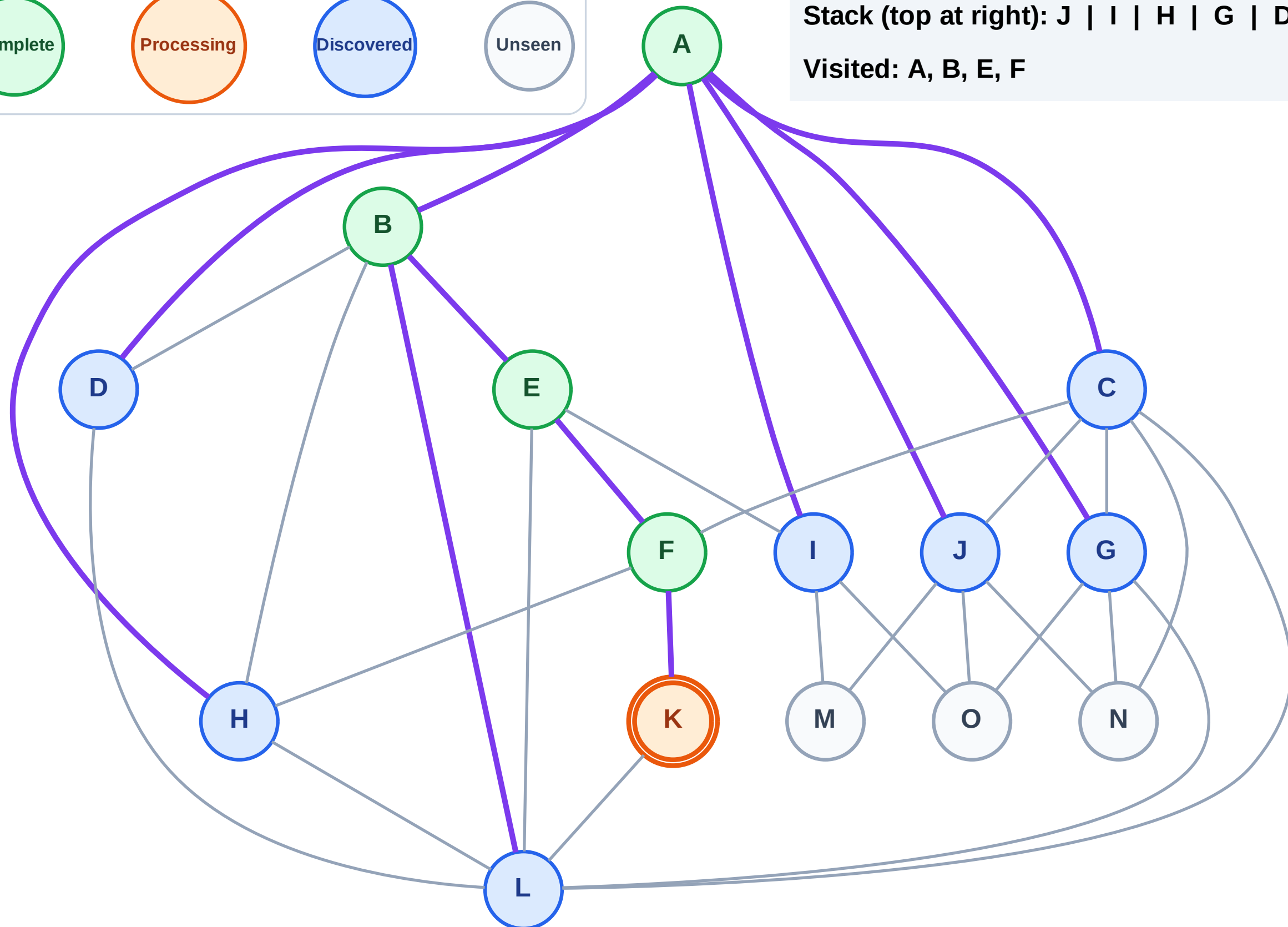
Discovered



Unseen

Stack (top at right): J | I | H | G | D | C | L

Visited: A, B, E, F



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (4 hops): A → B → E → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | H | G | D | C | L

Visited: A, B, E, F, K

